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Real-Time Speech Emotion Recognition Using CNN-BiGRU-Attention: A Study of Feature Richness, Architectural Complexity, and Live Deployment

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in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST)

DECLARATION

We hereby declare that the work presented in this Capstone Project is the result of our own investigation and efforts, conducted under the guidance and supervision of Mamun Ahmed, Associate Professor, Department of Computer Science and Engineering, Bangladesh Army International University of Science and Technology, Cumilla, Bangladesh. We confirm that no part of this project has been submitted, nor is being submitted, for the award of any other degree or diploma at any institution.

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APPROVAL

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ABSTRACT

Automatic speech emotion recognition is a challenging problem with diverse applications in healthcare, education, HCI (human-computer interaction), and customer service domains. In this work, we provide an end-to-end SER pipeline implemented by A to Z steps from data wrangling and feature engineering up to model training, error analysis, real participants assessment and live web deployment for predicting Emotion continuously from live audio. The hybrid dataset of 12,098 samples was composed of RAVDESS, TESS and CREMA-D and included real-world audio recordings collected from different television shows or movies for six emotion classes. We did feature extraction on low-level spectra (170-dimensional basic set and 269-dimensional advanced set including MFCC delta & delta-delta coefficients, chroma STFT, spectral contrast). We trained the four architectures (CNN-BiLSTM, CNN-BiGRU, CNN-BiGRU-Attention and MS-CNN-BiGRU-MD-Attention), on both feature sets. To measure model performance, a standard two-phase protocol was used to fine-tune and test against the SAVEE dataset for all models, while an additional diagnostic experiment was set up to ensure that any improvement in a model's behaviour could be attributed solely to feature richness rather than attention mechanism behaviour. The proposed model CNN-BiGRU-Attention with advanced features yielded 85.00 accuracy on SAVEE and 84 on real participant recordings, with fast inference latencies of an average of 43.11 ms, comparable to published results in the field. As a demonstration, the system was rolled out as a web app that conducted real-time emotion classification every 3 seconds. Results consistently show that features score more important than architecture complexity, and also show that how well a model is pre-trained isn't the best metric for its real-world usages.

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We are also thankful to all our teachers during our whole education, for exposing us to the beauty of learning. Finally, our deepest gratitude and love to my parents for their support, encouragement, and endless love.

LIST OF ABBREVIATIONS

SER	Speech Emotion Recognition
CNN	Convolutional Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
BiGRU	Bidirectional Gated Recurrent Unit
MS-CNN	Multi-Scale Convolutional Neural Network
MD-Attention	Multi-Dimensional Attention
MFCC	Mel-Frequency Cepstral Coefficients
ZCR	Zero Crossing Rate
RMS	Root Mean Square
STFT	Short-Time Fourier Transform
TTA	Test-Time Augmentation
NPY	NumPy Binary File Format
CSV	Comma-Separated Values
JWT	JSON Web Token
API	Application Programming Interface
FastAPI	Fast Application Programming Interface
JSON	JavaScript Object Notation
HTTP	Hypertext Transfer Protocol
CPU	Central Processing Unit
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song
TESS	Toronto Emotional Speech Set
CREMA-D	Crowd-Sourced Emotional Multimodal Actors Dataset
SAVEE	Surrey Audio-Visual Expressed Emotion
ReLU	Rectified Linear Unit

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CHAPTER 1

PROBLEM IDENTIFICATION & UNDERSTANDING

1.1 Introduction

The information content of human speech is many times higher than the decoding of the words. The emotional content of spoken utterances is conveyed in the tone, rhythm, pitch, and energy you hear and interpret without consciously thinking about it. This is now an important research challenge to replicate in machines known as Speech Emotion Recognition (SER), which involves the automatic detection and classification of human emotional states from audio signals. A key enabling technology, SER has proliferated into many different types of intelligent systems. Its applications span multiple domains:

- **Healthcare** : Detection of vocal indicators of stress, anxiety or depression for clinical intervention
- **Human-Computer Interaction** : How feedback can facilitate contextual and emotional responsiveness in virtual assistants and other dialogue systems
- **Education** : Identifying lack of engagement or confusion in online learning environments for adaptive teaching
- **Customer Service** : In real time, detect emotional unhappiness for better service quality and escalation handling

A basic constraint holds across nearly all previous SER research despite recent large advances from deep learning. Most models are trained and tested with acted emotional speech collected in well controlled laboratory settings. Despite being standardised and heavily used, these datasets fall short of capturing the spontaneous and contextually rich nature of emotional expressions relevant for deployment in practice. As a result, models that score high on benchmark evaluations often struggle to generalise across speakers, recording quality or unseen data.

The present study provides three key contributions that address these limitations. In the first study, we define a hybrid training dataset composed of three well-established benchmark corpora RAVDESS, TESS and CREMA-D and 1,200 novel crowd-sourced speech samples obtained

from various real-world media sources to infuse more acoustic diversity into the training dataset. Second, a CNN-BiGRU-Attention architecture is systematically compared against three alternative models with both basic 170 dimensional and advanced 269 dimensional sets of features to establish the controlled comparison between feature richness and architectural design. Third, beyond benchmark evaluation on both real and synthetic data, we perform an experiment with actual participants to validate the proposed model using newly captured natural multi-emotion signals while also measuring its inference latency for further benchmarking of its suitability for real-time deployment.

It also evaluates the model generalization to an unseen dataset (SAVEE). This implies that a model can explain training data very well, but in reality is completely silly and is repeated again and again, that given simple models (simple fitting functions) you have a 100% accuracy they will always do poor on new (test) data. In contrast, the proposed model in this work shows relatively stable performance which is better suited for real-world applications.

The project transitions from theory to practice, creating a web app in real-time emotion detection per 3 seconds live and conducting a study with participants using actual voice recordings. These practical components allow the system to be used, demonstrating its not only feasibility in experiments but its effectiveness with real world scenarios.

1.2 Research Motivation

This research motivation stemmed from the realization of a significant disparity between SER systems performance in laboratories and their usability in real-world environments. However, most of these studies report very high accuracy (often above 90%) because they test the model on a dataset it was trained on. However, when these models are implemented on separate data sets, disparate recording settings, or varied speakers their performance drastically declines. This illustrates a fundamental limitation: these types of models are unable to generalize well outside the environment in which they were operated.

1.2.1 The Generalization Gap in SER Research

- It is well known from the SER literature that in-distribution benchmark accuracy does not necessarily reflect performance in the real world. A great many studies report good

classification accuracy (>90 percent); however, these results are often derived from held-out samples taken from the same distribution as training data. However, performances degrade significantly when the same models are tested on unseen datasets, different speaker demographics, or varying recording conditions. The generalization gap. Direct impact on practical deployment:

- A health system that misclassifies emotional distress could cause a critical intervention to be postponed
- An erroneous emotion detection in an educational context can lead to inappropriate adaptive responses
- In customer service, missing frustration can cause problems to escalate unnoticed

1.2.2 Limitations of Existing Training Data

A second motivation relates to the character of datasets typically applied in SER analysis. Well-known benchmark corpora such as RAVDESS, TESS and CREMA-D are largely composed of acted speech recorded by trained actors in acoustically controlled settings. These resources offer the benefit of consistently labelled training data, but also have an unavoidable domain mismatch with naturalistic speech. Sincere emotional expression tends to be more nuanced, differential and contextualized than performed display. Training data lacking naturalistic samples thus restricts a model's access to the full spectrum of acoustic diversity present in spontaneous speech, and this limited exposure is a direct cause of poor cross-domain generalization. By including 1,200 real-world speech samples collected specifically for this study in a variety of noise conditions along with standard benchmark data, this study fills that gap.

1.2.3 Limitations of Basic Feature Representations

Feature selection is a key factor that affects the performance of SER models, but very few comparative studies quantify the influence of different features on generalization in the literature. While basic acoustic features (e.g., MFCC and RMS energy) faithfully represent spectral envelope characteristics, they likely do not encode the complete temporal and harmonic composition of natural emotional speech. Richer representations, or put more mathematically delta and delta-delta MFCC coefficients, Chroma STFT, Spectral Contrast have extra dimensions to describe how the acoustic features change over time.

1.2.4 Lack of Real-World Validation and Latency Benchmarking

This work is motivated by two additional gaps in the literature. The first one is that most of the SER studies end with benchmark evaluation and they do not evaluate their models on recordings by real participants in a naturalistic setting so it remains unclear if proposed systems would perform well in operating deployment scenarios. Second, almost all researchers know that inference efficiency is crucial for real-time applications, but only a few report latency measurements of their proposed architectures so that we can evaluate whether or not they are practically deployable. This study directly addresses both of these gaps by evaluating a participant with real subjects and benchmarking inference speed vs published systems.

1.3 Problem Statement

Although deep learning-based SER has made great progress, a long-standing well-known problem is that most systems perform well on controlled datasets but degrade substantially in real-world or cross-corpus conditions. There are several reasons for this limitation.

Root Causes:

Acted speech Bias in Data: The heavy reliance and dependence on acted benchmark datasets lead to models that are tuned for exaggerated, controlled emotional expression than actual spontaneous emotions present in real conversations. Introduces systematic bias, reducing generalisation to unseen speakers and recording conditions.

Architectural Constraints : Many traditional CNN-RNN structures suffer from limitation on a fixed feature dimension and lack of a mechanism to pay attention to emotionally salient temporal moments. When the attention component is not present, all time steps get equal treatment by the model and this is less than ideal because emotional cues are usually found in limited sections of an utterance.

Not Enough Understanding of Feature-Architecture Interactions: We systematically analyze how the choice of feature representation interacts with model architecture to impact both in-distribution accuracy and cross-corpus generalization. What works at a large scale often does not work at a smaller scale [14], and the interaction between feature richness and attention mechanism behaviour in small-scale fine-tuning is still underexplored.

Hypothesis (H1): A CNN-BiGRU-Attention architecture trained on a hybrid dataset combining acted and real-world speech will achieve strong emotion recognition performance across multiple emotion datasets, and training with advanced acoustic features including delta MFCC, Chroma STFT and Spectral Contrast will produce better recognition performance than basic features.

Null Hypothesis (H0): Architecture design and feature richness have no significant effect on Speech Emotion Recognition performance, a CNN-BiGRU-Attention model trained on advanced features will not perform significantly better than the same model trained on basic features across multiple emotion datasets and real participant recordings.

In order to tackle these issues, this project first introduces a new model architecture CNN–BiGRU-Attention, second uses combined acted and real-world data, and third does a detailed comparison of several models on both the custom and public datasets. The system is therefore more realistic and closer to the use case as it also incorporates a real-time application and contains a participant study of natural, mixed-emotion speech.

1.4 Research Objectives

The goal of this project is to accomplish the following:

O1. To create a hybrid speech emotion corpus pooling together three publicly available benchmark corpora RAVDESS, TESS, and CREMA-D with 1,200 newly hand-collected real-world speech data samples as well as systematically augmenting the data in order to increase acoustic diversity and the training volume.

O2. Allowing a controlled comparison of the effects of feature richness, we will design and extract two separate acoustic feature configurations, one consisting of 170 dimensions including MFCC, Log-Mel Spectrogram, ZCR and RMS energy (the baseline) and another containing 269 dimensionals made up of additional features: delta MFCC, delta-delta MFCC, Chroma STFT and Spectral Contrast.

O3. To deploy and assess 4 deep learning architectures (CNN-BiLSTM, CNN-BiGRU, CNN-BiGRU-Attention and MS-CNN-BiGRU-MDAttention) on both feature sets, identifying the CNN-BiGRU-Attention model as offering the best trade between in-distribution accuracy and inter-dataset generalisation.

O4. We train the CNN-BiGRU-Attention model twice, once with trainable attention and once with frozen (non-trainable) attention across each feature set to allow for a controlled diagnostic experiment that isolates the effect of richness of features vs behaviour of the attention mechanism on observed differences in performance.

O5. To apply our proposed CNN-BiGRU-Attention model on natural speech from non-native, naïve speakers containing multi-emotion with intentional transitions of emotions to measure real-world performance outside of traditional benchmark assessment.

O6. To evaluate the inference latency of the proposed SER systems under the consumer grade CPU hardware and validate against other published SER systems for its real-time applicability.

1.5 Research Questions

This project is driven by the following research questions:

RQ1. What is the effect of merging real-world spoken and acted speech datasets on the recognition performance of the CNN-BiGRU-Attention model when compared with benchmark datasets?

RQ2. Is the CNN-BiGRU-Attention architecture learning superior and more domain-agnostic features than relatively simpler CNN-RNN models, and what is its effect across simple and complex feature sets?

RQ3. In CNN-RNN based SER systems on our data, is there a sharp IN-OUT impressionistic trade-off between high in-distribution accuracy versus same protocol performance?

RQ4. Does CNN-BiGRU-Attention model perform well when we use natural and mixed emotions from recordings of real participants?

RQ5. Does more elaborate acoustic feature representation like delta MFCC, Chroma STFT and Spectral Contrast improve the recognition performance and generalization ability of the CNN-BiGRU-Attention model over basic features?

RQ6. Is the inference speed of the proposed CNN-BiGRU-Attention model sufficient for real-time speech emotion recognition and how does it compare with existing SER systems based on deep learning?

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Review

- **Chowdhury et al. (2025) [1]** Introduced a compact ensemble combining 1D-CNN and CNN-BiLSTM models that utilized MFCC, ZCR, RMSE, and Chroma STFT features. Their system demonstrated strong generalization, earning about 97–98% accuracy across several public datasets, and they stressed the importance of carefully designed augmentation to ensure stable cross-dataset behavior.
- **Cen et al. (2016) [2]** They constructed a real-time SER system with CNN and SVM classifiers using acoustic speech features such as MFCCs, pitch, and energy. Proposed for e-learning platforms, the technology monitored the emotions of students and had limitations regarding noise, mixed speaking styles, and maintaining accuracy under strict timing constraints.
- **Parvez et al. (2025) [3]** They developed a hybrid CNN–BiLSTM model by MFCCs and spectral descriptors for emotion recognition, aimed particularly at accessibility settings. Their model performed good on benchmark datasets but struggled with subtle emotions, varying speakers, and background disturbances.
- **Shukla et al. (2025) [4]** Proposed a CNN-based model that fuses MFCC and LPC information to improve emotional discrimination. Although accuracy was strong, the authors pointed out challenges involving dataset limitations, external noise, and the lack of robustness in real-world deployments.
- **Jena et al. (2025) [5]** Designed a deep CNN–LSTM solution for identifying negative emotional states such as anger, fear, and distress using MFCC and Log-Mel features. The approach performed reliably in risk-oriented environments, although class imbalance and unpredictable audio conditions remained significant obstacles.
- **Li et al. (2025) [6]** Introduced a Transformer-driven, model for low-latency SER that processes raw audio and spectrogram inputs. This method was suitable for interactive applications but needed further optimization toward noisy settings and inference time.

- **Kerkeni et al. (2019) [7]** Explored traditional machine-learning classifiers,SVM, Random Forest, and k-NN,using acoustic cues such as MFCCs, energy, and pitch. Their method worked for acted and spontaneous datasets in call-center and social media contexts, though generalization and noise tolerance remained limitations.
- **Hansen et al. (2022) [8]** Employed CNN, LSTM models together with prosodic and acoustic features (MFCC, Formants, Pitch), to identify depression symptoms. The result revealed significant diagnostic potential, but results were impacted by interspeaker variability, as well as imbalanced databases.
- **Grágeda et al. (2025) [9]** Proposed a CNN-LSTM hybrid solution that extracted MFCC, Chroma, and prosodic features for application in human robot interaction tasks. Although the solution operated in dynamic and static environments, it had difficulties in coping with varying noise conditions.
- **Zhou (2025) [10]** Proposed an emotion-aware speech diary tool that used CNN models with MFCC inputs for user emotion interpretation. In addition to proving the feasibility of the approach, the authors emphasized the importance of confidentiality in voice-based mood tracking systems.
- **Jegadeesan et al. (2025) [11]** Constructed a real-time SER system with a CNN+BiRNN architecture involving MFCC, spectral, and prosodic cues for mental health analysis. Even in real-time scenarios, accuracy remained high, implicitly supporting regular augmentation and personalized models.
- **Wu et al. (2024) [12]** Developed a benchmark called EMO-SUPERB, which tests multiple models within the SER category. They noticed discrepancies in the performance of different models, which supports the need for common evaluation practices within the discipline.
- **Chakhtouna et al. (2024) [13]** Had demonstrated the efficiency of self-supervised learning for Emotion Classification tasks, especially when leveraging Wav2Vec2, HuBERT models, because they produced a significant improvement with self-supervised learning-based features.

- **Gonçalves et al. (2024) [14]** Hosted the Odyssey 2024 SER Challenge, providing a set of datasets, a baseline, and evaluation procedures. The research revealed the challenges that exist in robustly performing SER on a set of real-world Emo-Categories and Audio conditions.
- **Wani et al. (2021) [15]** Presented a comprehensive overview on the state-of-the-art methods for SER, from classical solutions to deep learning solutions. The authors surveyed typical types of features, domains, and challenges that existing approaches face, such as inconsistency, noise, and low-resource scenarios.
- **Latif et al. (2023) [16]** Proposed a GAN framework to produce a synthesized emotional speech that generalized better for SER. Although useful, problems arose in the naturalness of the synthesized audio signal and how to properly use existing training samples.
- **Tang et al. (2025) [17]** Proposed a CNN + multi-attention mechanism in the Transformer architecture. The results showed high performance on multiple datasets, but computational cost and the need for attention mechanism tuning are still problems.
- **Barhoumi et al. (2024) [18]** Proposed a CNN-BiLSTM architecture reinforced with different augmentation techniques, including the addition of noise and changes in pitch/time. The architecture increased accuracy and robustness but lacked real-time performance.
- **Islam et al. (2024) [19]** Used a deep CNN architecture with MFCCs on spectrograms for emotion classification tasks, which worked quite well but was susceptible to overfitting when trained on small datasets, which led to a recommendation to use diversification techniques for augmentation.
- **Aftab et al. (2022) [20]** Introduced LIGHT-SERNET, a computationally efficient FCNN using spectrograms for SER. The model showed good accuracy with small size, but a trade-off between performance and efficiency was still challenging together with the corresponding methodology.
- **Hadžić et al. (2025) [21]** Introduced a hybrid method by combining model-based temporal with acoustic features (MFCCs, pitch and energy) to analyze emotional dynamics in semi-clinical interviews. They described difficulty in coping with variability of noise and psychological inconsistency.

- **Ouyang et al. (2025) [22]** Propose a CNN–LSTM model for MFCCs to trade off spectral and temporal learning, targeting interactive/education application. Drawbacks were the time delay of processing and the sensitivity to false background disturbances.
- **Alsabhan (2023) [23]** demonstrated 1D-CNN with attention and ensemble methods for real-time speech expression recognition; exhibited temporal and spectral feature extraction to increase performance of human–computer interaction systems; hurdles included dealing diverse noise levels as well as speech variation; recommendations noted that rich representations from integrating CNN in particular with Attention methods could be exploited for effective emotion classification.
- **Schuhmann et al. (2025) [24]** Built EmoNet-Voice, providing reliable assessment across emotions from different speakers; applications include benchmarking SER models and improving training datasets; challenges are annotation consistency as well as real-world conditions; emphasis on pairwise expert verification to achieve high-quality labels.
- **Abdelhamid (2023) [25]** Developed a speech emotion recognition system for online education using CNN and MFCC features, also displayed effective detection of student emotions, found challenges with noisy classroom data, and suggested feature optimization and model tuning for improved accuracy.
- **Chatterjee et al. (2021) [26]** Built a real-time speech emotion analysis system for smart home assistants using CNN + LSTM with MFCC and spectral features. They focused on more than one device applicability and robustness in domestic environments. Also found challenges in background noise and recommended attention-based improvements.
- **Bahreini et al. (2016) [27]** Proposed a real-time speech emotion recognition system for affective e-learning using GMM (Gaussian Mixture Model) with prosodic and spectral features; highlighted applications in online learning, addressed challenges like noise and latency, and suggested adaptive feature selection for better performance.
- **Zennou et al. (2026) [28]** CNN-BiLSTM-Attention architecture (MFCC-RMSE) for Real-Time SER on MFCC and RMSE features using Soft Attention Mechanism. Real-time compatibility was confirmed by an inference latency of 48 ms and throughput of 21 utterances/second on Intel Core i7 CPU, with a performance of 91.74% on RAVDESS, and 98.10% for the combination with RavTess.

Paper Name	**Architecture & Features**	**Reported Results**	**Limitations**
Chowdhury et al. (2025) [1] *Published At: 2025 Scientific Reports (Q1)*	1D-CNN + CNN-BiLSTM ensemble using MFCC, ZCR, RMSE, Chroma STFT on RAVDESS, TESS, CREMA-D acted datasets only, single fixed feature set	EmoDB 98.13%, RAVDESS 97.57%, TESS 100.00%, SAVEE 98.44%, CREMA-D 98.66%; every single result is a within-dataset split result	Only one fixed feature set used; no comparison of features; every result is within-dataset only; no performance check of models; no real-world data & real participant validation; no latency reported and no deployment
Barhoumi et al. (2024) [18] *Published At: 2024 Artificial Intelligence Review (Q1)*	CNN-BiLSTM with noise, pitch and time augmentation using MFCC and pitch features on RAVDESS and EmoDB, both fully acted datasets, single fixed feature set	≈95% accuracy on RAVDESS evaluated only within same dataset	Single feature set only; no feature richness analysis; no performance evaluation; no participant study; no latency measurement; no deployment
Zennou et al. (2026) [28] *Published At: 2026 Engineering Technology & Applied Science Research (Q2)*	CNN-BiLSTM-Attention using only MFCC and RMSE on RAVDESS and RavTessacted datasets, very limited feature set explicitly acknowledged as a limitation by authors	91.74% on RAVDESS / 98.10% on RavTess / 48 ms latency / 21 utterances per second one of very few papers to report latency but no cross-dataset evaluation	Only two features used; no delta or chroma features; no participant study; no feature analysis; no complexity analysis

Table 2.1: Overview of The Three Most Relevant Papers

2.2 Research Gaps

The literature review highlights a number of very specific, persistent gaps that demonstrate the direct relevance of this study. The gaps together drive the architecture choices and experimental design and validation strategy adopted in this work.

1. Limited Exploration of Rich Feature Representations in CNN-RNN Models

Though relatively large numbers of SER studies exist, many utilize a limited subset of acoustic features without systematically examining data representations to determine if more complex set predictions for improved generalization across datasets.

Chowdhury et al. [1], Parvez et al. [3], and Shukla et al. [4] all use common features like MFCC, ZCR and the RMS energy. Though these works have competitive results on benchmark datasets, none of them:

- Perform a controlled experiment of basic versus advanced feature configurations in an organized manner
- Decouple the influence of feature richness from architectural variances

Barhoumi and BenAyed [18] used a CNN-BiLSTM model layered with data augmentation techniques of background noise addition, pitch/time shifting to achieve better accuracy as well as robustness. However:

- There is no formal comparison of feature configurations in their study.
- We do not investigate the extent to which richer feature representations support cross-dataset transfer.

Zennou et al. [28] proposed a CNN-BiLSTM-Attention model that only uses two features (MFCC & RMSE) achieving decent results on RAVDESS and RavTess datasets. However:

- The study explicitly states its dependence on small feature types.
- Rich feature-based cross-domain performance restriction due to no evaluation on unseen datasets.

In this study, we fill in these gaps by creating and extracting two separate feature sets of audio , a simple 170-dimensional set; an advanced content derived from MFCC delta, delta-delta LFCC,

Chroma STFT and Spectral Contrast (totaling at 269 dimensions) features , then subject both datasets to all four architectures under consistent controlled conditions.

2. Limited Understanding of Attention Mechanisms in SER Models

Though attention mechanisms have largely been adopted in recent SER architectures, how exactly the behavior of various kinds of attention interacts with feature richness and cross-dataset generalization are still poorly understood out there based on existing literature.

Tang et al. [17] achieved superior accuracy across datasets with a CNN consisting of multi-attention Transformer architecture. However:

- With the advent of new architectures since then, however , namely Transformer encoder based ones that bring both high computational costs and large amounts of parameters to practically deploy.
- The study does not examine how attention behaviour evolves when the model encounters an alternative acoustic space.
- Explicit evaluation of cross-corpus generalization is missing.

Alsabhan [23] employs 1D-CNN based approaches with attention and ensemble strategies for real-time SER. However:

- This study neither looks at how different parts of the attention mechanism behave given inputs under basic vs advanced feature learning.
- Attention component is never truly tested in isolation on a diagnostic experiment.

The current study fills this gap with a controlled diagnostic experiment, training the CNN-BiGRU-Attention model from both trainable and frozen attention modes of each feature set while empirically isolating whether all observed cross-dataset performance disparities arise due to difference in feature richness or simply behavior of the mechanism within attention.

3. Lack of Naturalistic Data and Real-World Validation

Yet, equally concerning is the fact that almost all SER studies are created and tested only on a set of benchmark datasets with acted speech not integrated from naturalistic speech or validated using recordings made by participants in real world settings.

Chowdhury et al. [1], who explicitly state that the naturalistic and subjective nature of everyday

emotional expression is still an open problem to which acted corpora do not provide a solution, train and evaluate their system solely through publicly available but unvalidated datasets on displayed emotions.

Jena et al. [5], Grágeda et al. [9], and Ouyang et al. [22] all three of them target real-world application domains (namely mental health monitoring, human-robot interaction and education), but their models are built on a benchmark of acted dataset, validated only against that particular method. None of these studies:

- Gather or integrate naturalistic speech samples into training process.
- Do participant study to test model performance in realistic natural conditions.

The current study fills this gap in two respects. To begin with, 1200 speech samples of sincere emotion taken from real-world media television shows and movies are added to the training set along with standard benchmark data. Second, a separate participant study where real users give natural recordings of multiple simultaneous emotions along with planned emotion transitions form the basis of our validation.

4. Lack of Latency Evaluation for Real-Time Applications

One tangible yet frequently neglected gap in the SER literature is inference efficiency. Latency tolerance is critical to avoid deadlocks in real-time applications but not one of the studies discussed here provide measurements for latency or throughput within any applicable range.

Cen et al. [2], Chatterjee et al. [26], and Bahreini et al. [27] specifically mention that their research targets real time deployment scenarios but none report inference latency or throughput measurements to validate claims about being in-real-time.

Li et al. [6] introduced a Transformer-based low-latency SER model and mentioned that inference time needs to be lowered even further, but there are no explicit measurements on latencies on the work.

Zennou et al. [28] is one of the rare cases reporting average latency of 48 ms per utterance (on Intel Core i7 CPU) and achieving a frame rate of 21 utterances/second. However:

- Their analysis is confined to a particular architectural topology.
- In no case do we compare against other SER systems.

- Latency is NOT benchmarked in participant evaluation conditions.

The current study seeks to bridge this gap by measuring inference latency on consumer grade CPU hardware with real participant evaluation, and benchmarking against the results from Zennou et al. [28].

2.3 Research Contributions

This project makes the following key contributions to speech emotion recognition:

Contribution 1: A Hybrid Real-World and Acted Speech Dataset

- Manually collected 1,200 audio samples from real TV shows and films The Big Bang Theory, Modern Family, Breaking Bad, and The Conjuring where emotions occur naturally rather than being acted.
- Combined with RAVDESS, TESS, and CREMA-D to form a hybrid corpus of 12,098 samples across six emotion classes.

Contribution 2: The Proposed CNN-BiGRU-Attention Architecture

We design a hybrid deep learning architecture, by combining:

- Extracting spatial representations from acoustic features with CNN layers
- Bi-directional GRU (BiGRU) for speech temporal dependencies modeling
- Attention mechanism to focus on inputs corresponding to emotional segments

By replacing the traditional CNN-RNN models with unique BiGRU, we achieve lower computational complexity without hampering their strong sequence modeling powers. The attention layer helps improve the model focus on the regions of speech signal that give more information.

Contribution 3: Comparative Analysis of Feature Representations

Two distinct feature configurations are designed and evaluated:

- **Basic Features (NPY1)**: MFCC, Log-Mel Spectrogram, ZCR, RMS (170×300)
- **Advanced Features (NPY2)**: MFCC with delta and delta-delta, Log-Mel, Chroma, Spectral Contrast, ZCR, RMS (269×300)

Both feature sets are trained and evaluated exactly the same way on all four architectures. The results show that richer feature representations always lead to improved model robustness and generalization performance.

Contribution 4: Systematic Ablation Study Across Four Architectures

A structured architectural progression is designed and evaluated, consisting of:

- CNN–BiLSTM
- CNN–BiGRU
- CNN–BiGRU–Attention
- MS-CNN–BiGRU–MD-Attention

The results indicate that the CNN–BiGRU–Attention model provides a balanced performance-computational efficiency trade-off, making it the appropriate solution for real-world implementation.

Contribution 5: Controlled Diagnostic Experiment on Attention Behavior

- Our initial experiment is a controlled one, aiming to determine if the observed performance gap between basic and advanced feature sets originates from the attention mechanism instead of any other behavior exhibited by the learnt representations.
- With respect to attention, all attention configurations are applied on each feature set separately in model CNN-BiGRU-Attention while the model is trained with some trainable and some frozen attention.
- Performance across all four experimental conditions demonstrates that advanced features provide superior performance independent of attention configuration.
- This gives direct empirical evidence that the feature richness is the main factor responsible for performance improvement, not the behavior of attention mechanisms alone.

Contribution 6: Participant Study with Multi-Emotion Speech

- Natural speech recordings are gathered from real participants which consists of two/three emotions sequentially produced with an intentional 2-second gap between transitions.
- CSV file with ground truth emotion labels and timestamp for each segment.

- This represents a sliding window inference method, which allows the model to follow particular emotional changes of tracked speakers over time in a continuous stream of audio.
- Then the model is fine-tuned on these recordings from participants to teach it to recognize speech with multiple emotions in succession.

Contribution 7: Live Web-Based SER Application

We implement and deploy a web-based application that performs real-time emotion recognition from microphone input using the trained model. This demonstrates that the proposed architecture is not only more generalizable in controlled experiments but also practically deployable in a real interactive system.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Research Design

This study used an experimental design based on five stages: data collection, preprocessing and augmentation, feature extraction, model training and evaluation, and real-world deployment of the trained model. We created a hybrid dataset by merging three public corpora with a set of 1,200 audio samples collected in the real world by hand. To compare how feature richness affects performance, two feature sets were extracted from the same data,a simple and basic 170-d prosthetic dimension feature set and an advanced one of 269-d to see if it affects performance. We trained four deep learning architectures on both feature sets under the same conditions, fine-tuned and evaluated the best performing model on the SAVEE benchmark, and validated it on real participant recordings and implemented it as a live web application.

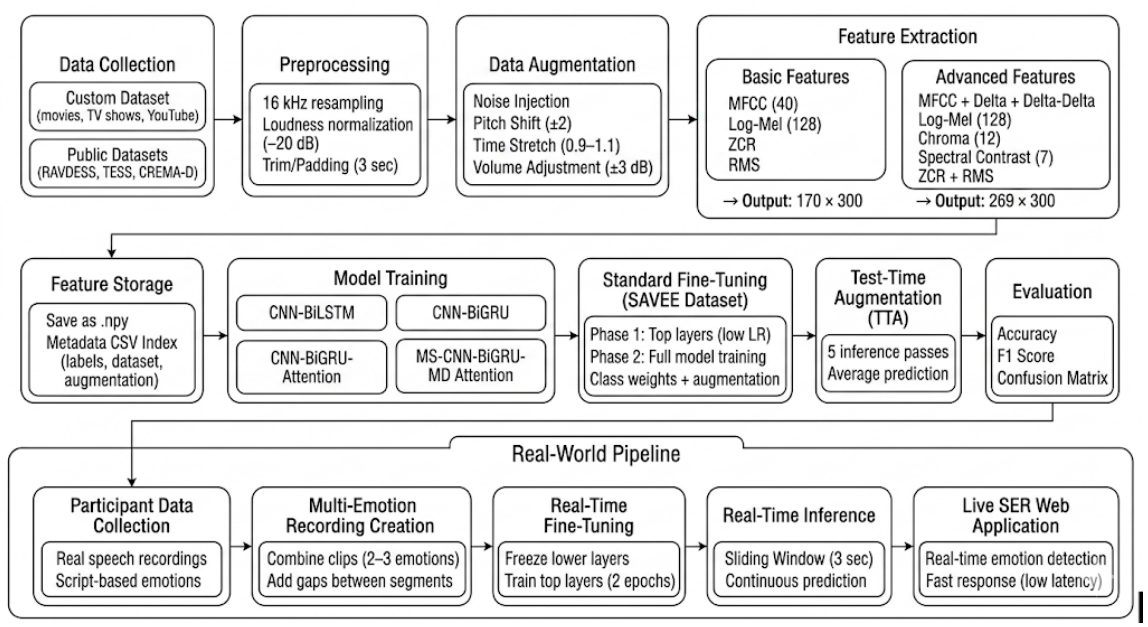


Figure 3.1: Research Design

3.2 Data Collection

3.2.1 Custom Data Collection

We made our dataset by watching a lot of television shows and films and clips from YouTube to see how people really show their emotions. We picked shows like The Big Bang Theory, American Housewife and Modern Family. Also Breaking Bad. We even looked at scenes from horror movies like Conjuring and videos of people investigating hotels. We listened to each clip carefully and used our own judgment to find the parts where people were clearly showing one of the six emotions we were looking for. When we found a part we downloaded the audio from a website, changed it to WAV format so everything was the same and then cut it down to a short clip that was 2.5 to 3.5 seconds long. We did this over and again until we had 1,200 clean audio samples with the same number of each of the six emotions. This way we could make a dataset that was focused and good quality that showed how people really express their emotions in everyday life.

3.2.2 Public Data Collection

We also used datasets that other people had already made to add variety and make sure we had a good range of emotions. These datasets were called RAVDESS, TESS and CREMA-D. We downloaded all of the datasets and then filtered them so we only kept the samples that showed the six emotions we were looking for. This helped us keep everything with our own custom data and it also added recordings from trained actors and people from different backgrounds. The result was a dataset that was balanced and reliable and that made our overall dataset more robust.

Dataset	**Happy**	**Sad**	**Angry**	**Neutral**	**Fear**	**Disgust**	**Total**
Custom	200	200	200	200	200	200	1200
RAVDESS	192	192	192	96	192	192	1056
TESS	400	400	400	400	400	400	2400
CREMA-D	1271	1271	1271	1087	1271	1271	7441
Overall Total	2063	2063	2063	1783	2063	2063	**12098**

Table 3.1: Data File Count Based on Emotion

3.3 Data Preprocessing

Once the custom audio samples collected by team members had been combined, all audio files (both from the custom dataset and the three public databases) were normalized following a standard procedure. We aimed to have a uniform sampling rate, loudness level and duration among the whole collection since it is key for reliable model training on speech emotion recognition tasks.

Monophonic, 16 kHz audio tracks were created by loading each audio file and then resampling the individual source at a common sampling rate. Loudness normalization was subsequently performed to adjust every sample so that its target level was consistent at -20 dBFS using root mean square (RMS) amplitude adjustment. Finally, all clips were made of exactly three seconds: files were padded with silence at the end if they are shorter, or contain only the first three seconds if longer. This preprocessing filtered out variations in recording length and quality that could interfere with the performance of our model.

3.4 Data Augmentation

To increase the size and diversity of the dataset and improve the model's ability to generalize, applied four types of audio augmentation techniques to the normalized files. These augmentations were applied one at a time to each original sample, resulting in four new variations per file.

The augmentation methods included:

- **Noise Injection**: This adds a small level Gaussian noise ($\alpha = 0.005$) to simulate background interference.
- **Pitch shifting**: Randomly shift the pitch ± 2 semitones to add some natural tone variation.
- **Time stretching**: Audio at some speed (rate between $0.9\times$ and $1.1\times$), but keeping the pitch, followed by duration correction.
- **Volume Change**: Randomly increase or decrease gain ± 3 dB, to account for different recording levels.

As such, it greatly augmented the dataset leading to a total of 60,490 audio samples being included in the overall data.

This enriched collection helps reduce overfitting and enhances the robustness of the speech emotion recognition model under varied acoustic conditions, as users may have noisy environments, bad microphone, user may talk very fast or slow, user may speak with a loud voice.

In addition, for the advanced feature set (269-dimensional NPY files), online feature-level augmentation was applied during model training. For each training batch, Gaussian noise was randomly added to feature values, and contiguous time frames or feature dimensions were randomly zeroed out using time masking and frequency masking respectively. Unlike offline augmentation, these operations are applied in memory during each training epoch, exposing the model to a slightly different version of each sample at every pass and further improving generalization.

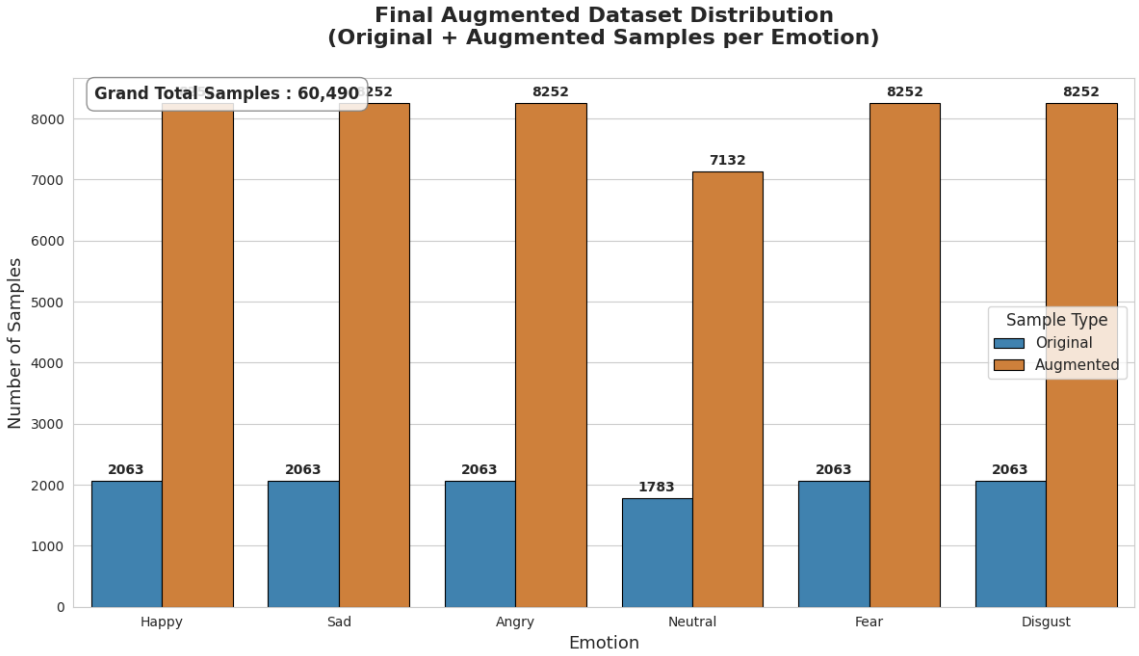


Figure 3.2: Emotion Distribution

Handling of Imbalance dataset:

The dataset used in this study consists of 12,098 speech samples across six emotion categories. While five classes Happy, Sad, Angry, Fear, and Disgust, each contain 2,063 samples, the Neutral class contains only 1,783 samples, making it the minority class.

Training on an imbalanced dataset without correction can bias the model toward majority classes, causing it to underperform on the Neutral category, one of the most commonly occurring emotional states in real-world speech.

To address this, class weights were computed using scikit-learn's `compute_class_weight` function with the 'balanced' setting, which assigns each class a weight inversely proportional to its frequency. As a result, Neutral received a weight of 1.131 while all other classes received 0.977. These weights were applied during training, causing the loss function to penalize Neutral misclassifications more heavily and effectively compensate for the lower sample count.

3.5 Feature Extraction and Storage

After completing the data augmentation process, we extracted acoustic features from all audio samples in the final augmented dataset. The goal was to convert the raw audio files into compact numerical representations suitable for training deep learning models for speech emotion recognition.

For every audio file, the following features were computed:

- **Mel-Frequency Cepstral Coefficients (MFCC)**: 40 coefficients per frame
- **Log-Mel Spectrogram**: 128 mel frequency bands, converted to decibel (dB) scale
- **Zero Crossing Rate (ZCR)**: calculated as the rate of sign changes in the audio signal
- **Root Mean Square (RMS) energy**: representing the energy envelope of the signal

To ensure consistent input dimensions across all samples, each feature matrix was fixed to a length of 300 time frames. This produced the following feature shapes for each audio sample:

- MFCC: 40×300
- Log-Mel Spectrogram: 128×300
- ZCR: 1×300
- RMS: 1×300

Each feature was standardized to a fixed length of 300 frames to ensure uniform input dimensions. The extracted features for every file were saved as a single `.npy` (NumPy) file containing a dictionary with all four feature arrays. This format (basic features) allows fast loading during model training while keeping the data efficiently organized.

Apart from this, we also extracted a second, more detailed set of features (advanced features) from the same augmented dataset. In this version, we added the delta and delta-delta derivatives of the MFCC coefficients to capture how the spectral characteristics change over time, along with Chroma STFT (12 features) and Spectral Contrast (7 bands). Each feature matrix was also fixed to 300 time frames, producing the following shapes per audio sample:

- MFCC (with delta and delta-delta): 120×300
- Log-Mel Spectrogram: 128×300
- Chroma STFT: 12×300
- Spectral Contrast: 7×300
- ZCR: 1×300
- RMS: 1×300

Stacking all of these together gives a final feature matrix of 269×300 for each sample. One important difference here is that we applied z-score normalization to each sample right at the time of extraction, so the saved features are already scaled and do not need any further preprocessing before training.

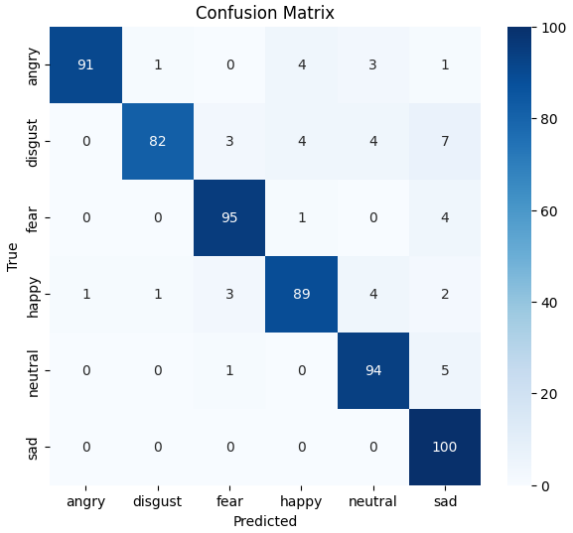
The feature extraction pipeline produced a well-organized, ready-to-use dataset that balances storage efficiency with rich acoustic information required for effective emotion classification. To maintain a clear record of the dataset, we also created a comprehensive index file. This CSV contains important metadata for every sample, including the original file name, emotion label, source dataset (Custom, RAVDESS, TESS, or CREMA-D), augmentation type (original, noise, pitch, stretch, or volume), and the relative path to the corresponding `.npy` file.

3.6 Quality Assessment of Custom Data

To verify the quality and usefulness of our manually collected custom dataset, we conducted a dedicated experiment using only the custom ("CUST") audio samples. This step was performed before training on the full combined dataset to evaluate whether the dialogues we selected and labeled through human judgment carried sufficiently clear and distinguishable emotional characteristics.

	precision	recall	f1-score	support
angry	0.99	0.91	0.95	100
disgust	0.98	0.82	0.89	100
fear	0.93	0.95	0.94	100
happy	0.91	0.89	0.90	100
neutral	0.90	0.94	0.92	100
sad	0.84	1.00	0.91	100
accuracy			0.92	600
macro avg	0.92	0.92	0.92	600
weighted avg	0.92	0.92	0.92	600

(a) Classification Report of Data Quality Assessment



(b) Classification Report of Data Quality Assessment

Figure 3.3: Quality Assessment of Custom Collected Data

CHAPTER 4

MODEL ARCHITECTURE AND EXPERIMENTAL SETUP

4.1 Overview of Proposed Model

This chapter presents the four deep learning architectures designed and evaluated in this study. All four models share a common foundation, a convolutional front end that extracts local acoustic patterns from the input feature sequence, followed by a bidirectional recurrent layer that captures temporal dependencies across the full utterance. The architectures differ in the type of recurrent unit used, whether an attention mechanism is included, and how the convolutional stage is structured.

The four models are:

- **CNN-BiLSTM**: uses a Bidirectional LSTM as the recurrent layer, making it capable of capturing long-range temporal dependencies in both directions across the utterance
- **CNN-BiGRU**: replaces the LSTM with a Bidirectional GRU, offering a simpler and more computationally efficient alternative while maintaining strong sequence modelling ability
- **CNN-BiGRU-Attention**: builds on CNN-BiGRU by adding an attention mechanism on top of the recurrent output, allowing the model to focus on the most emotionally relevant time steps in the sequence
- **MS-CNN-BiGRU-MD-Attention**: the most complex architecture, using multi-scale convolutional layers with parallel kernel sizes to capture patterns at different temporal resolutions, combined with a multi-dimensional attention mechanism

Each model was trained and evaluated on both the basic feature set (170-dimensional) and the advanced feature set (269-dimensional), giving eight configurations in total. All models were trained under identical conditions, same dataset, same augmentation pipeline, same class balancing strategy, so that any differences in performance reflect genuine differences between architectures and feature sets rather than variations in how the experiments were run.

Among all eight configurations, **CNN-BiGRU-Attention trained on advanced features** is the proposed model of this study. It achieved the best balance between pre-training accuracy and

fine-tuned performance, reaching 85.00% on the SAVEE benchmark and 84% on real participant recordings. The attention mechanism allows the model to focus on emotionally salient moments in the speech signal, while the advanced 269-dimensional feature set provides the rich spectral and temporal information needed for reliable performance.

4.2 CNN BiLSTM Architecture

The CNN–BiLSTM model integrates convolutional layers into a Bidirectional LSTM to learn local acoustic features and long-range temporal relationships in speech effectively. The input time–frequency features then go through a series of stacked Conv1D blocks with normalization, pooling and dropout leading to stable low-level emotional cues while suppressing noise in features at the same time minimizing dimensionality. A BiLSTM layer is then applied on top of the resulting feature sequence and learns the context from previous as well future time steps making it suitable for modeling how emotions progress over an utterance. Lastly, dense layers and a softmax classifier give us the output emotion prediction.

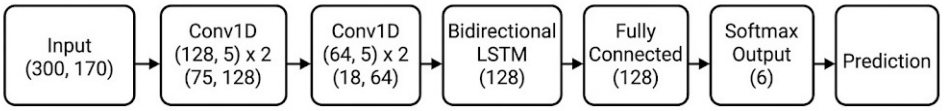
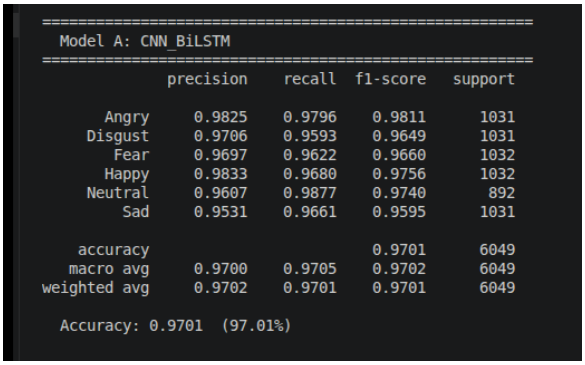


Figure 4.1: CNN BiLSTM Architecture

We have tested CNN-BiLSTM model on both basic features and advanced features set based on accuracy/standard classification metrics. For the simple features, it scored 97.01% accuracy and presents a good and stable performance on all emotion classes. It performed particularly well for Happy and Neutral with most errors occurring between close emotion categories such as Fear–Sad and Disgust–Angry, a finding in keeping with speech emotion recognition.

The drop to 94.28% accuracy on the advanced features range indicates the level of performance. Both versions achieved a similar stable performance across classes, however the confusion matrix showed a higher overall number of incorrect classifications for subtle emotions (Fear, Sad, Disgust), suggesting that the new features are not as discriminative compared to the older representations.



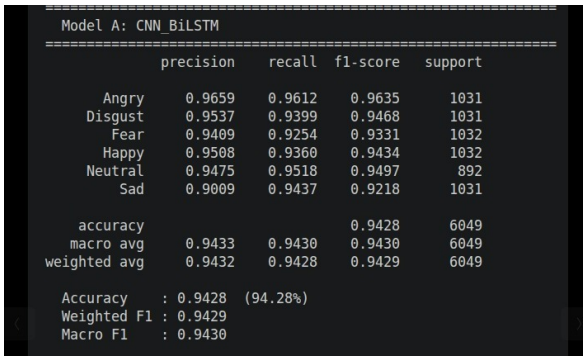
```
Model A: CNN_BiLSTM
              precision    recall  f1-score   support

       Angry     0.9825    0.9796    0.9811      1031
     Disgust     0.9706    0.9593    0.9649      1031
        Fear     0.9697    0.9622    0.9660      1032
       Happy     0.9833    0.9680    0.9756      1032
     Neutral     0.9607    0.9877    0.9740       892
         Sad     0.9531    0.9661    0.9595      1031

    accuracy                         0.9701      6049
   macro avg     0.9700    0.9705    0.9702      6049
weighted avg     0.9702    0.9701    0.9701      6049

Accuracy: 0.9701  (97.01%)
```

(a) CNN BiLSTM With Basic Feature



```
Model A: CNN_BiLSTM
              precision    recall  f1-score   support

       Angry     0.9659    0.9612    0.9635      1031
     Disgust     0.9537    0.9399    0.9468      1031
        Fear     0.9409    0.9254    0.9331      1032
       Happy     0.9508    0.9360    0.9434      1032
     Neutral     0.9475    0.9518    0.9497       892
         Sad     0.9009    0.9437    0.9218      1031

    accuracy                         0.9428      6049
   macro avg     0.9433    0.9430    0.9430      6049
weighted avg     0.9432    0.9428    0.9429      6049

Accuracy    : 0.9428  (94.28%)
Weighted F1 : 0.9429
Macro F1    : 0.9430
```

(b) CNN BiLSTM With Advance Feature

Figure 4.2: CNN BiLSTM Classification Report

4.3 CNN BiGRU Architecture

Similarly, the CNN–BiGRU model has a similar feature extraction stage based on Conv1D layers that automatically learn local speech patterns from time–frequency features while reducing noise and complexity in progressive steps. A Bidirectional GRU model is preferred instead of the LSTM to simultaneously fit temporal dependencies in both directions and as well a more parsimonious method with less computational cost. It will then feed through a softmax layer that outputs the final emotion class probabilities after being passed via dense layers (with regularization).

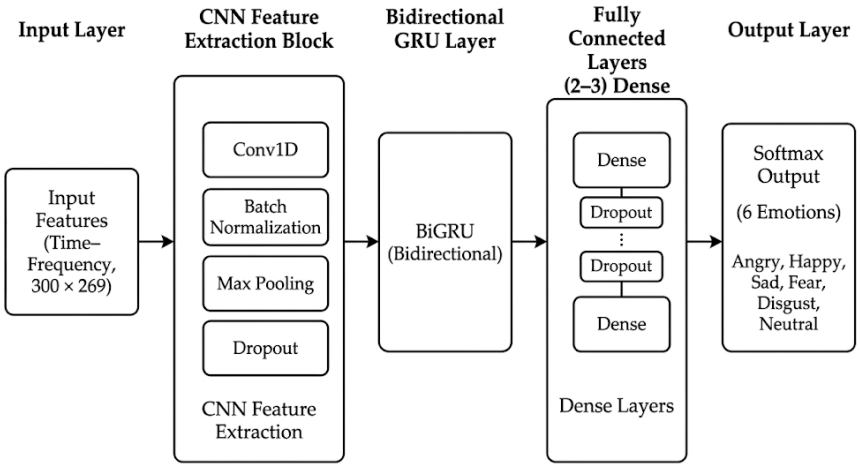
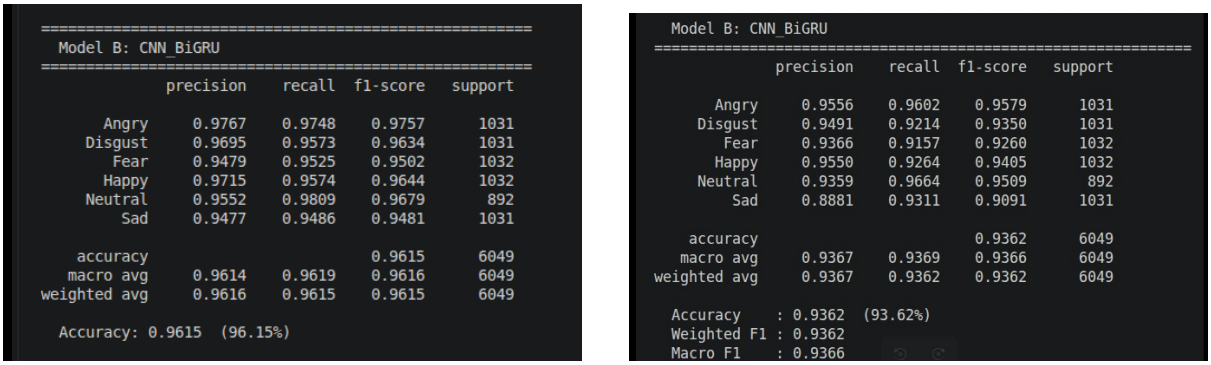


Figure 4.3: CNN BiGRU Architecture

The CNN–BiGRU model was tested for old and new NPY feature sets using accuracy and standard classification metrics. It reached 96.15% accuracy with all the basic features, performing consistently over most classes but slightly behind CNN–BiLSTM in general and closely related emotion categories.

The accuracy reduced further to 93.62% with the increased confusion in Fear, Sad and Disgust due to its advanced features. However, the model still maintains more or less stable overall performance suggesting it can generalise reasonably well even with a representation of advanced features.



Model B: CNN_BiGRU

	precision	recall	f1-score	support
Angry	0.9767	0.9748	0.9757	1031
Disgust	0.9695	0.9573	0.9634	1031
Fear	0.9479	0.9525	0.9502	1032
Happy	0.9715	0.9574	0.9644	1032
Neutral	0.9552	0.9809	0.9679	892
Sad	0.9477	0.9486	0.9481	1031
accuracy			0.9615	6049
macro avg	0.9614	0.9619	0.9616	6049
weighted avg	0.9616	0.9615	0.9615	6049

Accuracy: 0.9615 (96.15%)

(a) CNN BiGRU With Basic Feature

Model B: CNN_BiGRU

	precision	recall	f1-score	support
Angry	0.9556	0.9602	0.9579	1031
Disgust	0.9491	0.9214	0.9350	1031
Fear	0.9366	0.9157	0.9260	1032
Happy	0.9550	0.9264	0.9405	1032
Neutral	0.9359	0.9664	0.9509	892
Sad	0.8881	0.9311	0.9091	1031
accuracy			0.9362	6049
macro avg	0.9367	0.9369	0.9366	6049
weighted avg	0.9367	0.9362	0.9362	6049

Accuracy : 0.9362 (93.62%)
Weighted F1 : 0.9362
Macro F1 : 0.9366

(b) CNN BiGRU With Advance Feature

Figure 4.4: CNN BiGRU Classification Report

4.4 CNN BiGRU Attention Architecture

This CNN–BiGRU–Attention model combines convolutional feature extraction, bidirectional recurrent modeling with an attention mechanism for speech emotion recognition. Input time–frequency features are fed into stacked Conv1D layers, learning local spectral-temporal patterns and eliminating noise while pooling/normalization reduces size of the feature map. Next, a Bidirectional GRU layer takes in the resulting sequences of features to capture both forward and backward temporal dependencies with lower model complexity than LSTM. We apply an attention layer on top of the recurrent output to focus only on those time steps in the sequence that are most relevant for emotion detection. Finally, the output from three fully connected dense layers with dropout regularization is passed through the softmax layer for emotion classification.

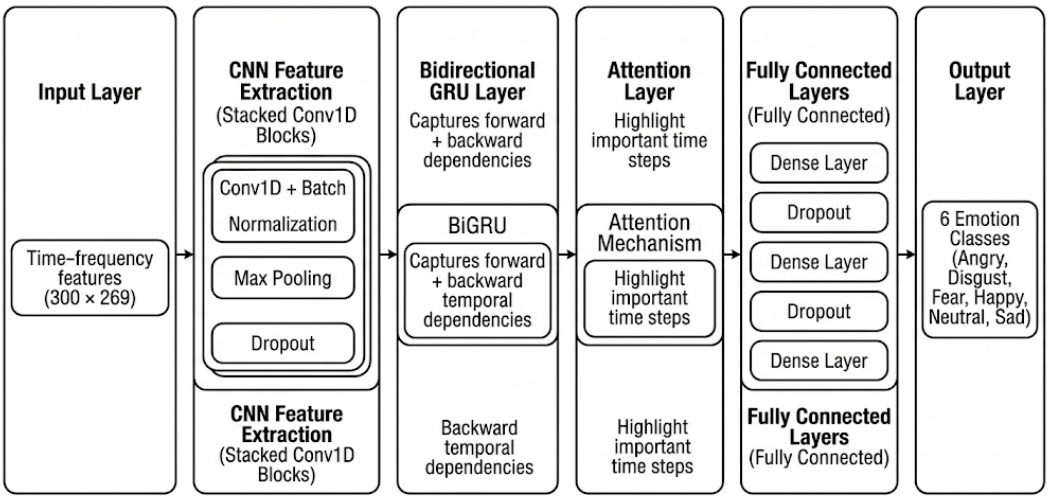


Figure 4.5: CNN BiGRU Attention Architecture

The model is performing well with an accuracy of about 94% so far on the basic features sets. Although most emotions were reliably classified, the best agreement was achieved with Happy followed by Neutral and Angry. As speech emotion recognition is characterized by overlapping acoustic patterns, it can be observed that most errors occurred between similar emotions (e.g. Fear–Sad and Disgust–Angry).

For those with advanced feature set, the model was able to achieve a weighted F1-score of 0.9311 and accuracy of 93.12%. All classes remain stable, with slight increases in confusion between closely related emotions: Happy–Angry and Fear-Sad being the worst performers.

	precision	recall	f1-score	support
Happy	0.9601	0.9564	0.9582	1031
Sad	0.9586	0.9428	0.9506	1031
Angry	0.9792	0.9583	0.9687	1032
Fear	0.9221	0.9632	0.9422	1032
Disgust	0.9591	0.9545	0.9568	1032
Neutral	0.9675	0.9697	0.9686	891
accuracy			0.9572	6049
macro avg	0.9578	0.9575	0.9575	6049
weighted avg	0.9575	0.9572	0.9572	6049

Accuracy : 0.9572 (95.72%)
Weighted F1 : 0.9572
Macro F1 : 0.9575

(a) CNN BiGRU Att With Basic Feature

CNN-BiGRU-Attention — Combined

	precision	recall	f1-score	support
Angry	0.9013	0.9738	0.9361	1031
Disgust	0.9388	0.9224	0.9305	1031
Fear	0.9421	0.9147	0.9282	1032
Happy	0.9704	0.8895	0.9282	1032
Neutral	0.9135	0.9709	0.9413	892
Sad	0.9259	0.9214	0.9237	1031
accuracy			0.9312	6049
macro avg	0.9320	0.9321	0.9313	6049
weighted avg	0.9324	0.9312	0.9311	6049

Accuracy : 0.9312 (93.12%)
Weighted F1 : 0.9311
Macro F1 : 0.9313

(b) CNN BiGRU Att With Advanced Feature

Figure 4.6: CNN BiGRU Attention Classification Report

4.5 MSCNN-BiGRU with Multi-Dimensional Attention Architecture

Multi-Scale CNN–BiGRU with Multi-Dimensional Attention (MS-CNN-MBA) Model aims to represent informative emotional patterns in audio signals. The multi-scale Conv1D layers were designed with multiple kernel sizes (3, 5 & 7) in order to extract complementary temporal features enabling the model on learning both short- and long-term acoustic variations.

A Bidirectional GRU processes these features in order to learn dependencies across the whole input sequence. We then apply a multi-dimensional attention mechanism to highlight the most informative time steps and feature channels, enabling improved focus on emotionally relevant patterns by our model. At last, the attended features are pooled and fed to a softmax classifier for emotion prediction through dense layers.

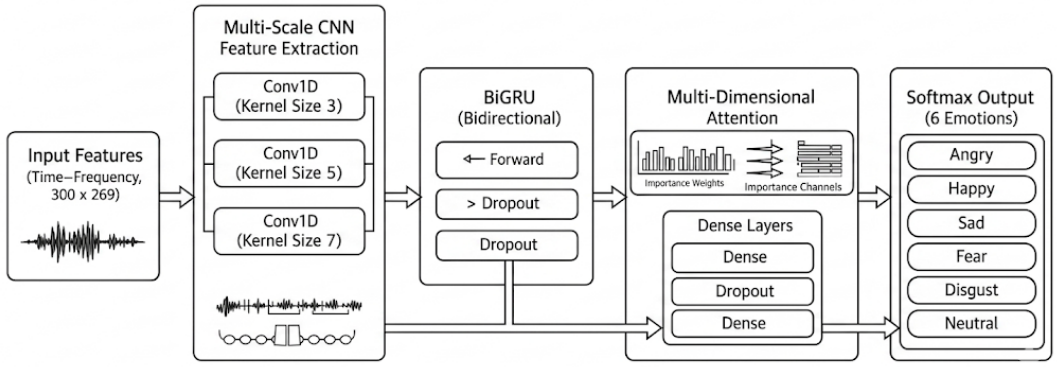


Figure 4.7: Multi-Scale CNN-BiGRU with Multi-Dimensional Attention Architecture

With the basic features set, it reached an accuracy of 92.81% and a weighted F1-score equal to 0.9281 on this model. The model was able to obtain pretty good performance on the majority of emotion classes. And this model worked particularly well for the emotions Angry, Fear and Neutral.

The model obtained a much improved performance on the advanced features set with an accuracy of 96.43% and weighted F1-score equal to 0.9643. All emotion classes have a high precision and recall especially Angry, Disgust and Neutral, and this thing demonstrates good feature discrimination.

MultiScale-CNN-BiGRU-MultiDimAttn — Combined

	precision	recall	f1-score	support
Happy	0.9911	0.8681	0.9255	1031
Sad	0.8678	0.9486	0.9064	1031
Angry	0.9258	0.9671	0.9460	1032
Fear	0.8944	0.9438	0.9184	1032
Disgust	0.9794	0.8769	0.9254	1032
Neutral	0.9310	0.9697	0.9500	891
accuracy			0.9281	6049
macro avg	0.9316	0.9290	0.9286	6049
weighted avg	0.9316	0.9281	0.9281	6049

Accuracy : 0.9281 (92.81%)
Weighted F1 : 0.9281
Macro F1 : 0.9286

(a) MSCNN BiGRU MD-ATTENTION With Basic Feature

	precision	recall	f1-score	support
Angry	0.9787	0.9787	0.9787	1031
Disgust	0.9699	0.9690	0.9694	1031
Fear	0.9523	0.9486	0.9505	1032
Happy	0.9888	0.9419	0.9648	1032
Neutral	0.9722	0.9787	0.9754	892
Sad	0.9277	0.9709	0.9488	1031
accuracy			0.9643	6049
macro avg	0.9649	0.9646	0.9646	6049
weighted avg	0.9648	0.9643	0.9643	6049

Accuracy : 0.9643 (96.43%)
Weighted F1 : 0.9643
Macro F1 : 0.9646

(b) MSCNN BiGRU MD-ATTENTION With Advanced Feature

Figure 4.8: MSCNN BiGRU MD-Attention Classification Report

CHAPTER 5

PERFORMANCE EVALUATION AND ANALYSIS

5.1 Test Dataset

The SAVEE database contains 480 utterances recorded by four native male British English speakers, with each speaker contributing 120 sentences across seven emotion categories. The data is evenly distributed, with 15 utterances per emotion per speaker, giving 60 samples per emotion across the full dataset. The seven categories covered are anger, disgust, fear, happiness, sadness, surprise, and neutral. For this project, the surprise category was excluded to maintain consistency with our six-class emotion scheme, resulting in a final test set of 420 samples with 60 utterances per emotion.

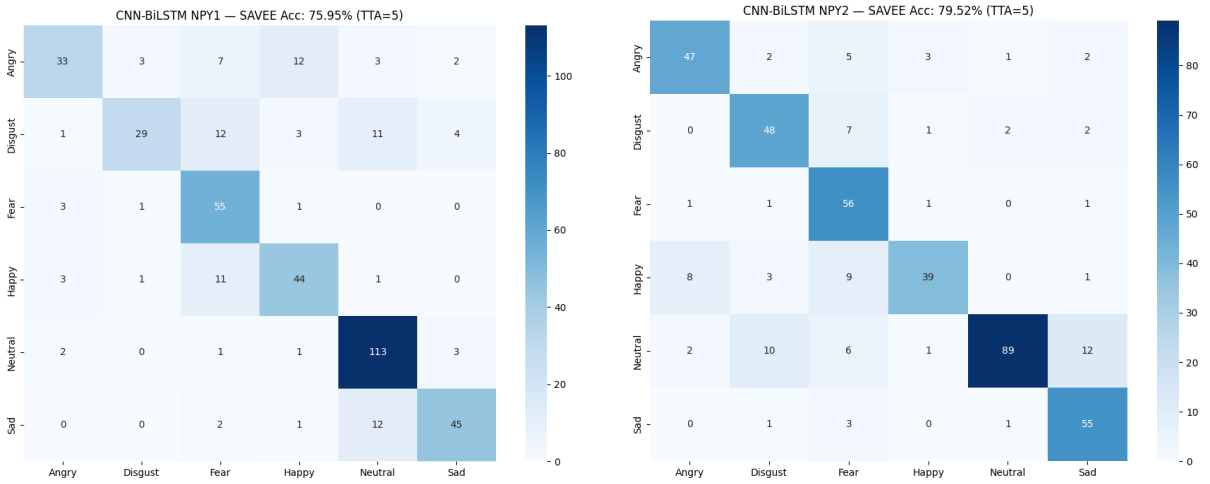
SAVEE was kept entirely unseen during training and was used purely as a standardised benchmark to compare all model configurations under the same conditions. Rather than testing cross-corpus generalisation to a different domain, the purpose here was to evaluate each model on a clean, balanced, and consistent test set that none of the models had been exposed to, ensuring that the comparison between architectures and feature sets is fair and unbiased.

5.2 Fine-Tuning Protocol and Evaluation

To evaluate how well each model configuration performs on an unseen dataset under a consistent, fair protocol, all models were fine-tuned on the SAVEE dataset before evaluation. This fine-tuning was applied uniformly across all eight configurations so that the comparison remains controlled.

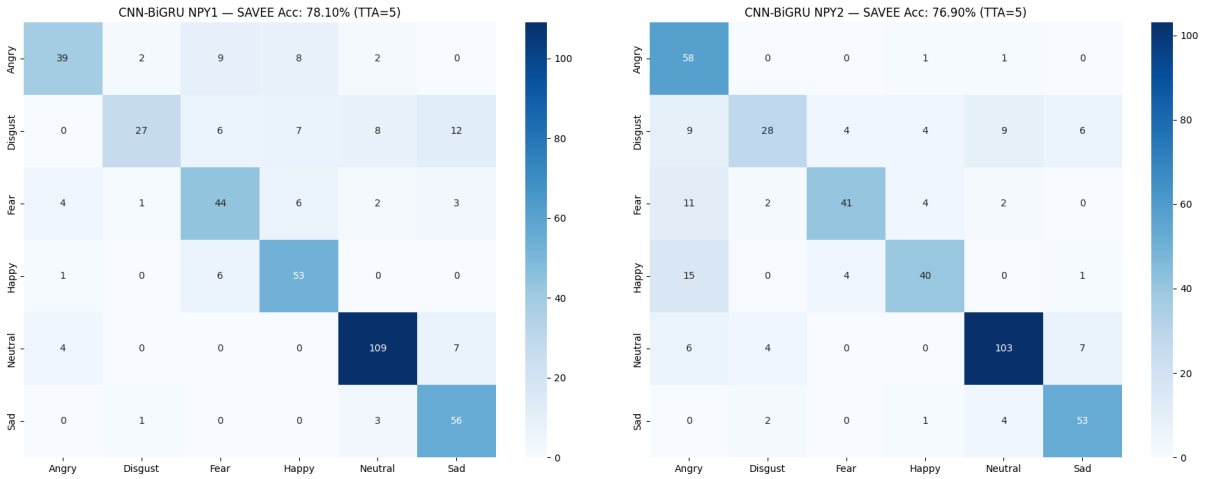
- **Phase 1 (Top layer warm-up)**: In the first phase, only the upper layers of the network were unfrozen for training. These layers were trained for 8 epochs using a low learning rate of 1×10^{-5} with gradient clipping set to a norm of 1.0. The lower convolutional layers were kept frozen during this phase, preserving the acoustic representations built up during pre-training while allowing the model to begin adjusting its higher-level decision boundaries to the SAVEE data.

- **Phase 2 (Full fine-tuning)**: In the second phase, all layers were unfrozen and the entire network was trained together for up to 24 epochs using learning rate of 5×10^{-5}. A learning rate scheduler was also applied to reduce the learning rate by half if no improvement was seen after 4 consecutive epochs, with a minimum floor of 1×10^{-8}. Class-balanced weights were applied throughout both phases.
- **Test-time augmentation (TTA)**: At evaluation time, five inference passes were run for each sample , one clean pass and four passes with stochastic augmentation applied. The softmax probability outputs from all five passes were averaged to produce the final prediction.



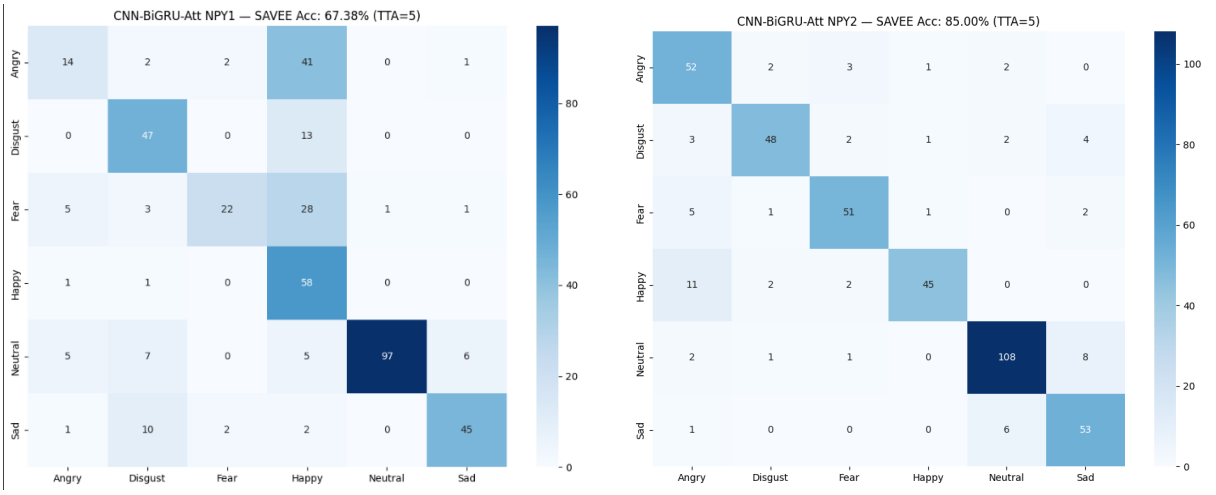
(a) CNN-BiLSTM NPY1 (b) CNN-BiLSTM NPY2

Figure 5.1: CNN-BiLSTM SAVEE Confusion Matrices



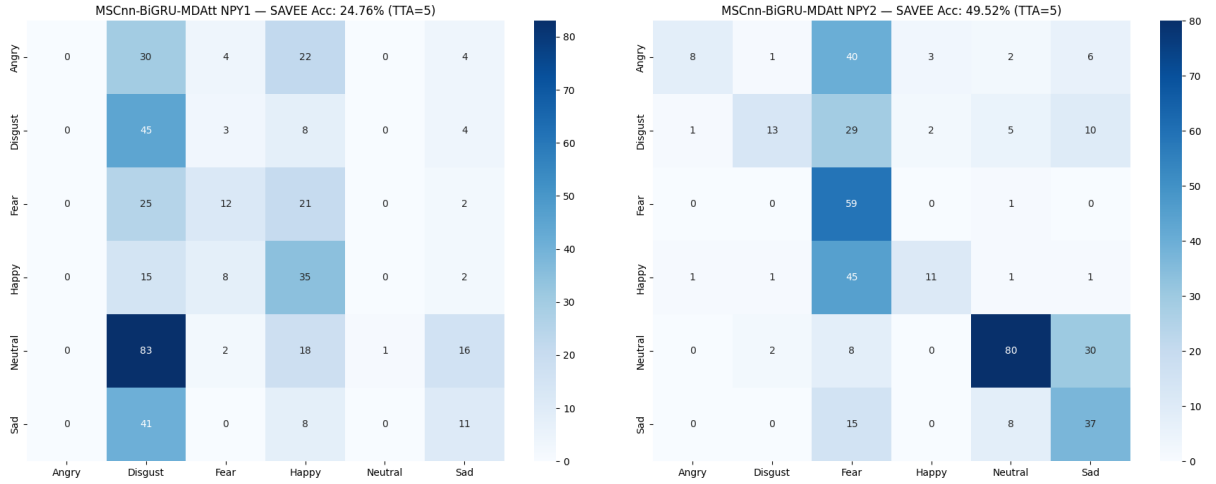
(a) CNN-BiGRU NPY1 (b) CNN-BiGRU NPY2

Figure 5.2: CNN-BiGRU SAVEE Confusion Matrices



(a) CNN-BiGRU-Attention NPY1 (b) CNN-BiGRU-Attention NPY2

Figure 5.3: CNN-BiGRU-Attention SAVEE Confusion Matrices



(a) MSCNN-BiGRU-MDATT NPY1 (b) MSCNN-BiGRU-MDATT NPY2

Figure 5.4: MSCNN-BiGRU-MDATT SAVEE Confusion Matrices

The confusion matrices tell a clear story most mistakes happen between emotions that naturally sound similar, like Fear and Sad, or Disgust and Angry, which is exactly what you would expect. Models using advanced features tend to get more predictions right across the board, with stronger diagonal patterns visible in their matrices. The biggest difference shows up in CNN-BiGRU-Attention, where switching to advanced features visibly cleans up the confusion matrix which makes sense given the 17 percentage point jump in accuracy between the two.

5.3 Overall Performance Results

The table below shows the pre-training accuracy and the fine-tuned SAVEE accuracy for all eight configurations across both feature sets.

Model	Basic Features		Advanced Features	
	Accuracy	**SAVEE (FT)**	**Accuracy**	**SAVEE (FT)**
CNN-BiLSTM	97.01%	75.95%	94.28%	79.52%
CNN-BiGRU	96.15%	78.10%	93.62%	76.90%
CNN-BiGRU-Attention	95.72%	67.38%	93.12%	**85.00%**
MS_CNN-BiGRU-MD_Attention	92.81%	24.76%	96.43%	49.52%

Table 5.1: Overall Performance Results

Starting with the pre-training results, all models do very well, every configuration scores above 92% regardless of which feature set is used. This tells us that all four architectures can learn the emotion recognition task well when the training and testing data come from the same source. But when we look at the fine-tuned SAVEE results, things look quite different. The scores now range from 24.76% all the way up to 85.00%.

- **Advanced features tend to give better results after fine-tuning**: In three out of four models which are CNN-BiLSTM, CNN-BiGRU, MS-CNN-BiGRU-MD-Attention, using advanced features leads to a higher or equal the SAVEE accuracy compared to the basic features. The biggest improvement is in the CNN-BiGRU-Attention, which goes from 67.38% with the basic features up to 85.00% with the advanced features a gain of more than 17 percentage points just by changing the feature set.
- **CNN-BiGRU goes against this trend**: The CNN-BiGRU is the only model where the basic features (78.10%) slightly outperform the advanced features (76.90%) after doing fine-tuning. The difference is small but it suggests that adding more feature dimensions does not always help, especially when the amount of fine-tuning data is limited.

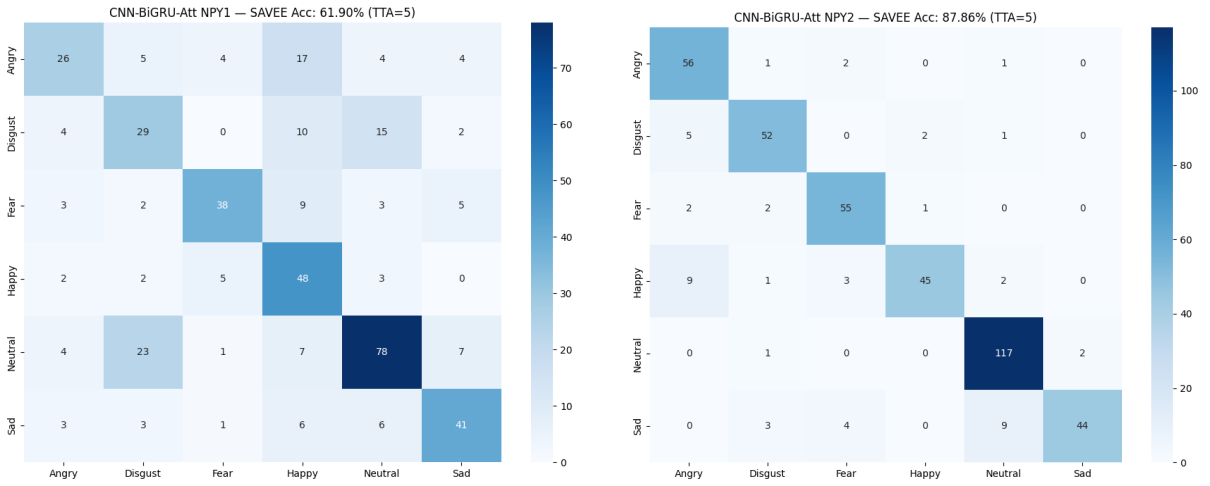
- **MS-CNN-BiGRU-MD-Attention has the most dramatic drop**: This model goes from 92.81% during pre-training down to just 24.76% after fine-tuning on SAVEE with basic features. On a six-class task, that is barely above guessing randomly. Even with advanced features it only reaches 49.52%, which is still the lowest result in the table.

5.4 Diagnostic Analysis

Two results stood out as unusual and needed a closer look CNN-BiGRU-Attention dropping to 67.38% with basic features despite training well, and MS-CNN-BiGRU-MD-Attention nearly collapsing to chance level after fine-tuning. Both results were unexpected enough that simply reporting the numbers without investigating further would have left an important question unanswered. So rather than moving on, we ran additional experiments to understand what was actually causing each drop.

5.4.1 CNN-BiGRU-Attention Control Experiment

Two results stood out and needed a closer look — CNN-BiGRU-Attention dropping sharply with basic features, and MS-CNN-BiGRU-MD-Attention nearly collapsing after fine-tuning. Rather than just reporting the numbers, we investigated both to understand what was actually causing them.



(a) CNN-BiGRU-ATT-NPY1-FREEZ (b) CNN-BiGRU-ATT-NPY2-FREEZ

Figure 5.5: Attention-Frozen CNN-BiGRU-ATT SAVEE Confusion Matrices

Model variant	Feature set	Pre-train accuracy	SAVEE (FT)
CNN-BiGRU-Attention	Basic (170)	95.72%	67.38%
CNN-BiGRU-Attention	Advanced (269)	93.12%	85.00%
CNN-BiGRU-Att-Freez	Basic (170)	95.47%	61.90%
CNN-BiGRU-Att-Freez	Advanced (269)	92.21%	87.86%

Table 5.2: Attention-Freezing Experiment

Three observations follow from this table:

- **Attention is not the cause of the drop**: The frozen-attention variant scores 61.90% on basic features , slightly lower than the original 67.38%, not higher. Both variants produce a similarly weak result, which means the attention layer is not responsible for the degradation.
- **Advanced features improve performance regardless of attention handling**: The original model scores 85.00% with advanced features while the frozen-attention variant scores 87.86% , a small gain of about 2 percentage points. This suggests that attention weights which were not locked into source-domain patterns during pre-training may adapt slightly more freely during fine-tuning.
- **The attention layer contributes a small benefit with basic features**: The difference between 67.38% and 61.90% shows the attention layer does help slightly when it is free to update from its pre-trained state.

5.4.2 MS-CNN-BiGRU-MD-Attention Result

The MS-CNN model drops from 92.81% pre-training accuracy to just 24.76% after fine-tuning with basic features , the most severe drop across all eight configurations. On a six-class task this is close to chance level. With advanced features the same model recovers to 49.52%, which is an improvement but still the lowest fine-tuned result in the table.

- **The drop is too large to be normal.** A fall of nearly 68 percentage points suggests the model is fundamentally struggling to adapt to SAVEE.

- **The multi-scale design may be the cause.** Using three parallel convolutional filters with kernel sizes 3, 5, and 7 creates a large number of parameters that become very specialised to the training data. With only 420 SAVEE samples available for fine-tuning, there is simply not enough data to readjust all of these parameters.
- **Advanced features help but do not fix the problem.** The gain from 24.76% to 49.52% shows richer features reduce the gap, but the model still finishes last overall.

5.5 Real Participant Evaluation

We tested CNN-BiGRU-Attention with advanced features on audio collected from our own participants. This was done to get a sense of how the model behaves on real, natural, variable length audio which contains 2 to 3 emotions.

Data collection from participants:

- Each participant was given a script and asked to speak with a specific target emotion.
- They recorded a 10-second audio clip for each of the six emotion categories.
- We also created longer combined recordings by joining two or three emotion clips together with a 2-second gap between them, to test whether the model could track emotion changes within a single continuous audio stream.
- Ground truth labels were saved in a CSV file alongside each recording so accuracy could be computed automatically during evaluation.

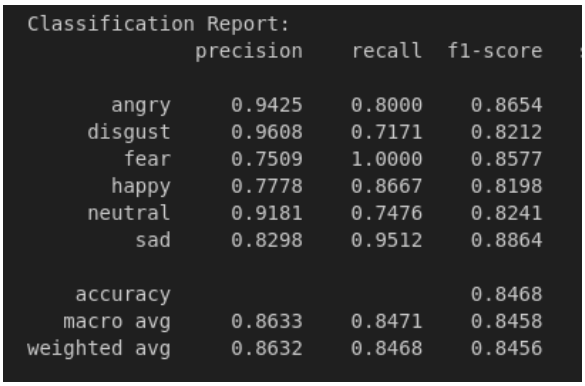
Feature extraction and fine-tuning:

- Features were extracted using the same advanced feature pipeline used throughout the study.
- During fine-tuning on participant data, only the upper layers of the model were updated for up to 2 epochs with a learning rate of 1×10^{-4}.
- The lower layers were kept frozen to avoid overfitting given the small size of the participant dataset.

Inference and results:

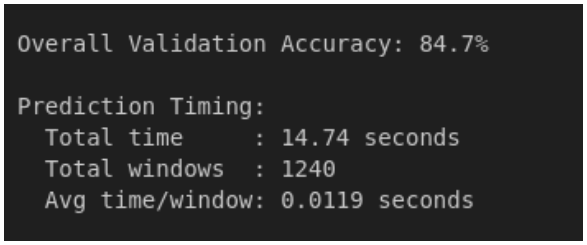
- A sliding window of 3 seconds was applied at inference time, producing one prediction every 3 seconds across the full recording.
- The model achieved an accuracy of **84%** on participant recordings.
- The average prediction time was **0.0119 seconds** per window, which is fast enough for real-time use.

These results show that the model performs well not just on standard benchmark data but also on naturally recorded speech from real participants.



Classification Report:			
	precision	recall	f1-score
angry	0.9425	0.8000	0.8654
disgust	0.9608	0.7171	0.8212
fear	0.7509	1.0000	0.8577
happy	0.7778	0.8667	0.8198
neutral	0.9181	0.7476	0.8241
sad	0.8298	0.9512	0.8864
accuracy			0.8468
macro avg	0.8633	0.8471	0.8458
weighted avg	0.8632	0.8468	0.8456

(a) Multi-emotion Fine-tune



```
Overall Validation Accuracy: 84.7%

Prediction Timing:
  Total time     : 14.74 seconds
  Total windows  : 1240
  Avg time/window: 0.0119 seconds
```

(b) Prediction Time

Figure 5.6: Real Participant Evaluation

5.6 Inference Latency

Beyond accuracy, a practical real-time system also needs to be fast enough to keep up with incoming audio. To measure this, we ran a dedicated latency benchmark on the best performing model , CNN-BiGRU-Attention with advanced features.

Benchmark setup:

- A single dummy input of shape $(1, 300, 269)$ was used to simulate one 3-second audio window.
- One warm-up prediction was run first to load the model into memory before timing began.
- 300 consecutive predictions were then timed using wall-clock time.

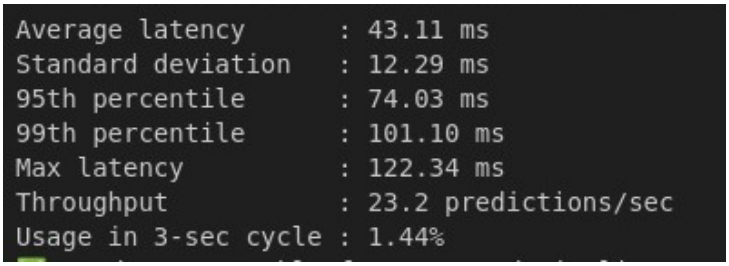


Figure 5.7: Model Latency Test

Key observations:

- On average each prediction takes 43.11 ms, which is well within the 3000 ms window available between predictions.
- Even in the worst recorded case of 122.34 ms, the prediction step uses only about 4% of the available cycle time.
- The 99th percentile latency of 101.10 ms means that in 99 out of 100 predictions the model responds in under 110 ms.
- The model only consumes 1.44% of the 3-second cycle on average, leaving the remaining time free for audio capture and feature extraction.
- A throughput of 23.2 predictions per second means the model could theoretically support much faster prediction intervals than 3 seconds if needed.

These results confirm that the model is comfortably fast enough for a real-time 3-second emotion prediction pipeline.

CHAPTER 6

RESULT AND DISCUSSION

6.1 Role of Feature Richness

The core observation: In most configurations, switching from basic features (170-dimensional) to advanced features (269-dimensional) improved fine-tuned accuracy on SAVEE. The biggest gain was in CNN-BiGRU-Attention, which jumped from 67.38% to 85.00% an improvement of over 17 percentage points just by changing the feature set.

Why do advanced features transfer better?

- Basic features capture a snapshot of the speech signal at each moment what the spectrum looks like, how much energy is present, how often the signal crosses zero. These are useful but sensitive to differences between speakers and recording environments.
- Advanced features add MFCC delta and delta-delta coefficients on top of the basic set. These capture how the spectral characteristics are changing over time. Emotional speech is dynamic, and the way a voice rises, falls, and shifts shape carries important emotional information that basic features miss. Chroma and spectral contrast add further tonal and harmonic information.

The one exception (CNN-BiGRU):

- CNN-BiGRU performs slightly better with basic features (78.10%) than with advanced features (76.90%).
- The difference is small but suggests that not every architecture benefits equally from higher dimensional input.
- With only 420 samples available for fine-tuning, a model receiving 269-dimensional input has more to process and adjust than one receiving 170-dimensional input.

Key takeaway: Feature engineering matters more than architectural complexity when it comes to adapting to new data. Investing in richer, more informative features is likely to give better returns than making the model architecture more elaborate.

6.2 Attention Mechanism Analysis

The core observation: CNN-BiGRU-Attention achieved 95.72% during pre-training with basic features but only 67.38% after fine-tuning on the same protocol, a drop of nearly 28 percentage points.

What the control experiment found:

- The frozen-attention variant scored 61.90% on basic features, slightly lower than the original 67.38%, not higher.
- With advanced features, the frozen-attention variant scored 87.86% compared to 85.00% from the original, a small improvement of about 2 percentage points.
- Overall, changing how the attention layer behaves produced no dramatic difference in either direction.

Three things this tells us:

- **Attention is not the cause of the drop.** If the attention mechanism were responsible for the 67.38% result, changing it should have produced a noticeably different outcome.
- **Attention is not a decisive factor with advanced features either.** Both variants perform well, 85.00% and 87.86% respectively.
- **Attention helps slightly at the margin with basic features.** The difference between 67.38% and 61.90% shows the attention layer does contribute something useful.

What this means for model design: The control experiment makes one thing clear, the feature set is the decisive factor here, not the attention mechanism. Good features matter more than attention layer design when it comes to adapting to new data.

6.3 Model Complexity and Adaptability

The core observation: MS-CNN-BiGRU-MD-Attention is the most complex model in the study and also the one that adapted the least successfully to SAVEE. It dropped from 92.81% pre-training accuracy to just 24.76% after fine-tuning with basic features.

Why did this happen?

- **Complexity is an advantage during pre-training.** The multi-scale convolutional design captures patterns at multiple temporal resolutions simultaneously using parallel kernel sizes of 3, 5, and 7.
- **The same complexity becomes a problem at fine-tuning time.** A model that has learned very detailed and specific patterns from its training data has more to unlearn and readjust when it encounters new data.
- **420 samples is not enough to recalibrate a complex model.** The simpler models have fewer parameters to adjust and recover more easily with the same amount of fine-tuning data.

What this means practically:

- A more powerful architecture does not automatically mean better real-world performance.
- When fine-tuning data is limited, a simpler well-featured model will likely outperform a complex one.
- Model selection for fine-tuning tasks should always take into account how much adaptation data is available.

6.4 Real-World Performance and Latency

The core observation: The best performing model CNN-BiGRU-Attention with advanced features achieved 85.00% on SAVEE after fine-tuning and 84% on our own participant recordings.

Participant results in context:

- SAVEE consists of acted speech from trained male British English speakers recorded under controlled studio conditions.
- Our participant recordings were made in less controlled environments by people who were not professional actors.
- Despite this, the model achieved a comparable accuracy of 84%, very close to the 85.00% SAVEE result.

- The combined recording test confirmed the model can track changing emotions across a continuous stream.

Latency results:

Two latency measurements were recorded:

- **During participant evaluation** , the model ran continuously in an active pipeline and got an average latency which is 11.9 ms per prediction window.
- **Dedicated benchmark** , where at first we run a proper warm-up, after that 300 isolated prediction calls were timed which gives an average latency of 43.11 ms with a 99th percentile of 101.10 ms.

6.5 Limitations

While the results are promising, there are a few honest limitations worth acknowledging which are given below:

- **SAVEE is a speech from a narrow speaker group.** All four speakers from this dataset are native male British English speakers performing scripted emotions in a controlled environment.
- **The fine-tuning dataset is small.** With only 420 samples across the six classes which are happy, sad, angry, neutral, disgust, fear, each class has only 70 samples on average for fine-tuning.
- **The participant dataset is limited.** Recordings were collected from a small number of participants under semi-controlled conditions.
- **The control experiment used the full SAVEE set.** There was no separate held-out split within SAVEE for the control experiment.
- **Latency was measured on a single machine.** The benchmark results reflect the hardware used during this study.

6.6 Summary of Findings

Three findings stand out consistently across all the experiments in this study:

- **Feature richness is the most important factor for performance of a model**: Advanced features outperformed basic features in most configurations, and no architectural improvement compensated for a weak feature set.
- **Pre-training accuracy is not a reliable measure of real-world performance.** All models trained well, but their performance results varied enormously.
- **Simpler models adapt more reliably when fine-tuning data is limited.** The most complex model in the study produced the weakest fine-tuned results while the simpler architectures generalised more consistently.

CHAPTER 7

REAL TIME LIVE SPEECH EMOTION RECOGNITION

7.1 System Overview

These theoretical contributions and experimental findings explored in the previous chapters coalesce here into a fully-fledged, real-time Speech Emotion Recognition (SER) web application that has been designed and deployed. Although most research systems are still limited to offline experiments, we aimed to show that the proposed CNN-BiGRU-Attention model is both more accurate and applicable in a real time implementation or live system through this web application.

At first the system takes an audio input directly from the user's microphone and then passes it through a trained deep learning model, for every three seconds, we get the emotion prediction in almost real-time. Not only are users can see the detected emotion but also by logging in, existing users are able to see the history of their emotional trends, receive analytical summaries and usage insights.

It structures itself in a client-server format and is based on the following three main technologies: a Next.js front-end app that directs the interface and audio capture, a FastAPI back-end service that manages model inference and business logic, and a MongoDB database to persist user data and detections.

7.2 System Architecture

Figure 7.1 shows architecture of the deployed system. Their design revolves around two separate usage flows , a public detection flow, which is accessible without logging in, and an authenticated flow for creating permanent features; however both utilize the same core inference pipeline at the back end.

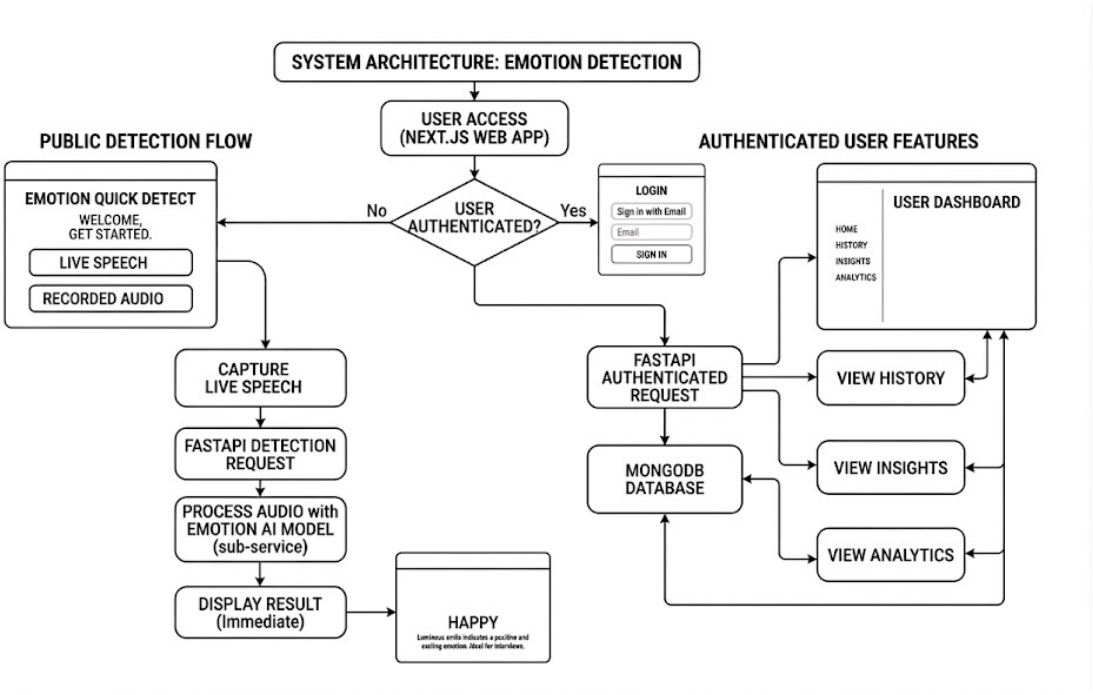


Figure 7.1: Emotion Detection System Architecture

The Next.js web application is a single point of entry (for all users). First, the system branches based on whether the user is authenticated. Users who are not authenticated will be taken to the open detection facility, where they can immediately start to capture and break down speech without logging in. Authenticated users, on the other hand, are routed through the FastAPI authentication layer and end up seeing a personal dashboard.

For both flows, the public flow and the authenticated flow steer down to the same emotion inference sub-service which runs as a different containerized process covered under trained CNN-BiGRU-Attention model. Authenticated sessions also cause a write operation to the MongoDB database in order that all detection results are stored permanently on the user account.

7.3 Front-End Application

The frontend was built on Next.js , a React Framework for developing fast, smooth and mobile first experience with server-side rendering. There are two main functions that the front end needs to perform: recording the user's microphone and displaying emotion detection results in a readable and interpretable format.

UI/UX Design Considerations:

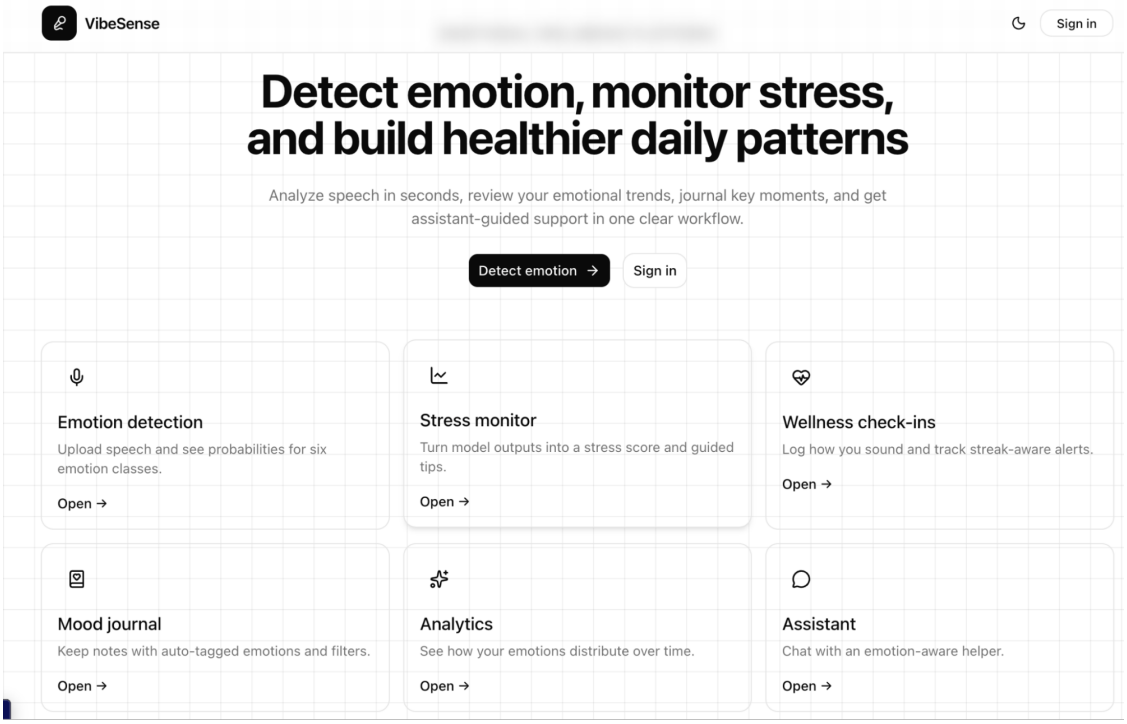


Figure 7.2: Home View & Navigation

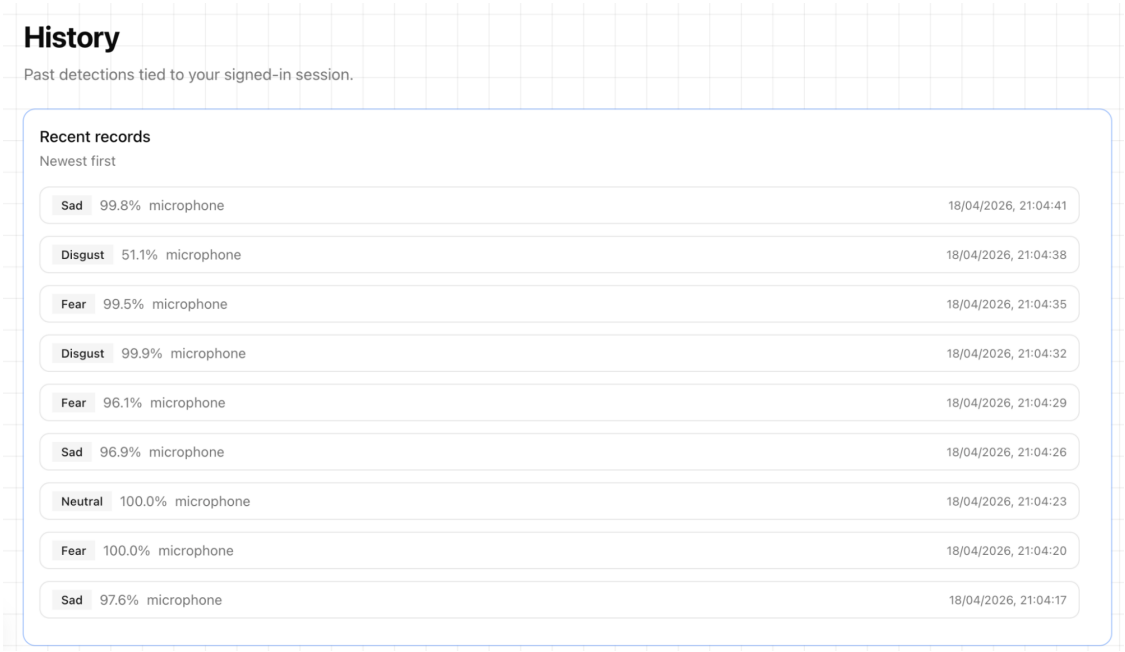


Figure 7.3: Historical Records & Logs

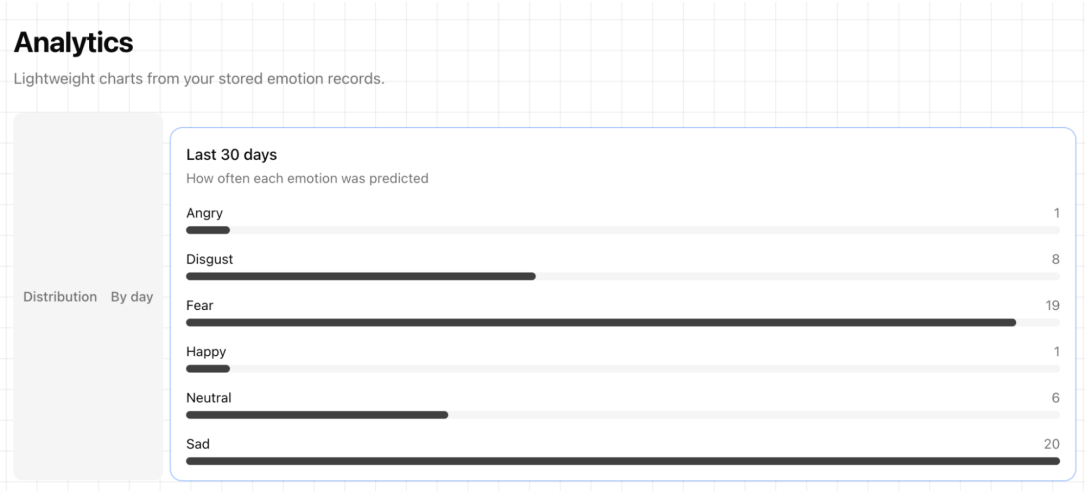


Figure 7.4: Statistical Data & Analytics

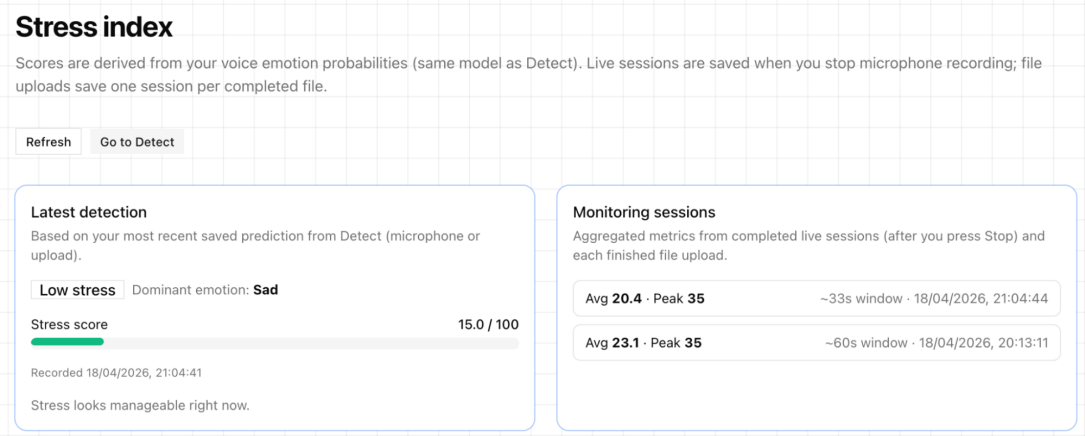


Figure 7.5: Stress Assessment Overview

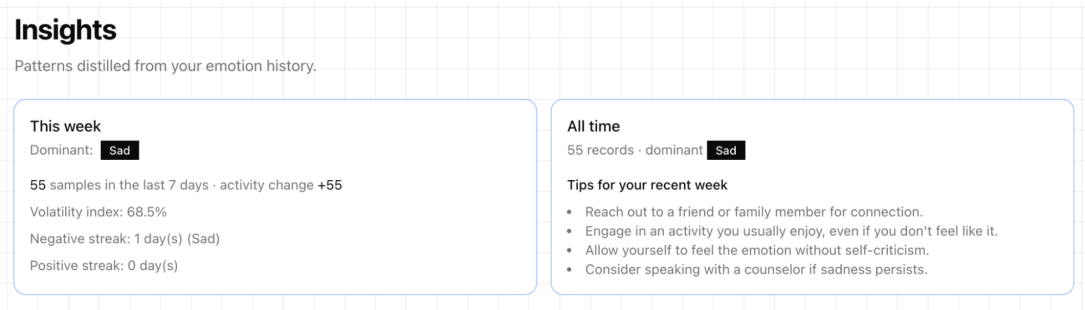


Figure 7.6: Behavioral Insights & Patterns

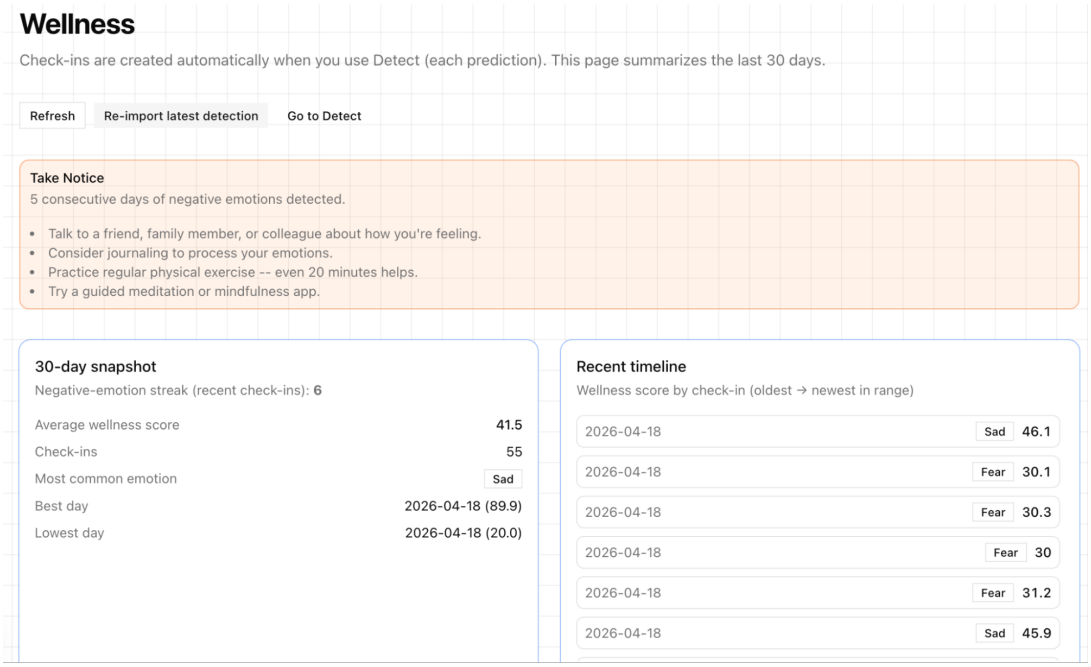


Figure 7.7: Wellness & Support Resources

7.3.1 Audio Capture and Streaming

If a user tries to start a live detection session, the browser makes an access request for the device microphone (as usual Web Audio API). After the user gives permission, the system goes in to record audio nonstop in the background. Instead of allowing the user to finish with their phrase and then examining a long recording, the app segments an audio stream into static three second windows. After every window, the recorded audio chunk is transmitted to the back-end server as a binary-formatted payload in the form of a WAV file via HTTP POST request.

This windowed approach parallels the three-second sliding window we used during model training and participant evaluation to ensure that inference conditions in the live system matched those under which we optimized our models. The three second interval was chosen on practical grounds , it allows sufficient temporal context for reliable emotion classification while avoiding an observable delay from the user perspective.

7.3.2 User Interface and Feedback Display

The public-facing interface, called Emotion Quick Detect, offers two input modes for the user: live speech detection via a mic or uploaded recorded audio (for offline use). If using it for live sessions, every three seconds being processed will be drawn out as an emotion on your screen and a short descriptive text message.

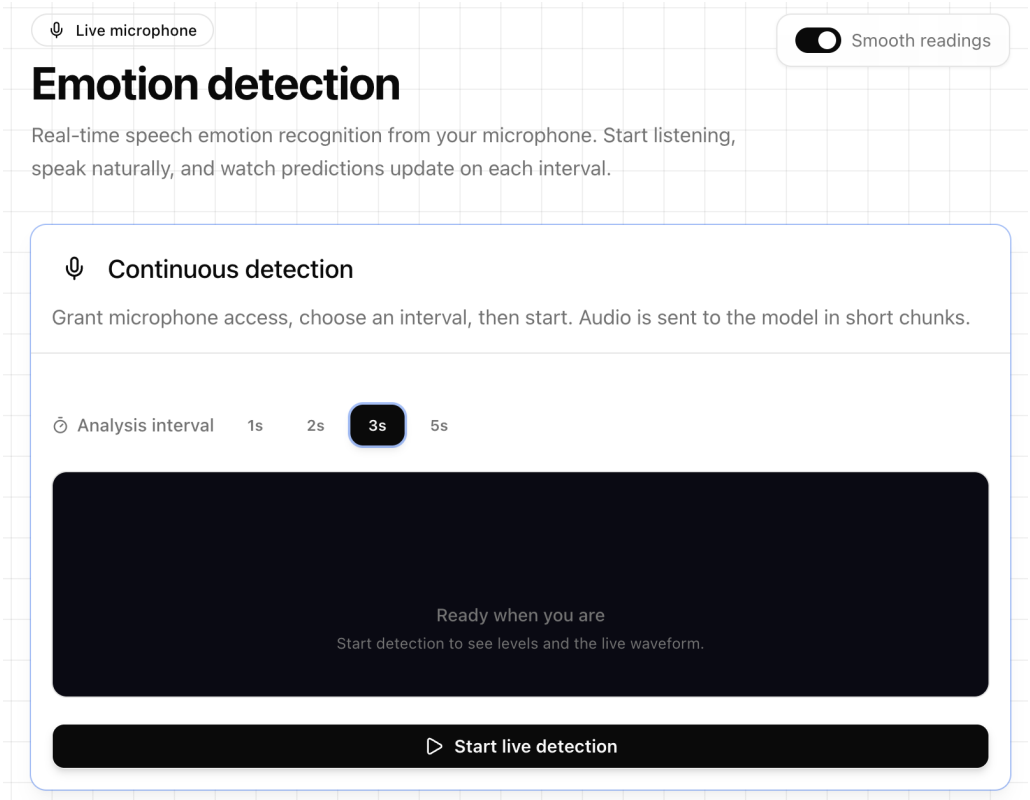


Figure 7.8: Continuous Emotion Detection

The interface for authenticated users opens into a standardized dashboard broken down by four navigation sections , home, history, insights and analytics.

7.4 Back-End Service

FastAPI is a non-blocking asynchronous API Python web framework, permits the service handling different requests with high speeds, formed the back end layer of this system. It was chosen compared to Flask or Django due to native support for asynchronous request handling.

7.4.1 Detection Endpoint

The core endpoint of back-end service receives incoming audio data from front end, it then does the pre-processing to make input format identical for the trained model and sends through inference pipeline. The pre-processing applied for this stage is exactly the same as the one we used at training time: resample to 16 kHz, convert to mono, normalise loudness at -20 dBFS, and pad/truncate to three seconds. The advanced feature set is then extracted and entered into the model.

It proceeds to output a probability distribution over the six emotion classes, from which we take the predicted emotion and its confidence score. The entire response payload , the predicted label followed by a confidence value , is served back to the front end as a JSON object and rendered in terms of the browser usually under 1 second from when this audio chunk was sent.

7.4.2 Authentication with JWT

JSON Web Tokens (JWT): The FastAPI back end checks the information against records before a user authenticates by logging in and, if confirmed, returns a signed JWT for authentication. Tokens are safely stored in browsers and added to header calls for every single request after that.

Server-side, protected API routes check the token for validity and ensure its signature and expiry prior to processing the request. We chose this stateless authentication mechanism as it removes the need for server-side session storage, allowing easy horizontal scalability.

7.4.3 Data Logging

The simple flow that any authenticated user provides an audio chunk to analyze: the back-end service calls the same inference steps as the public flow and then writes a record in a structured way in MongoDB. Each entry in the log includes the detected emotion label, confidence score, timestamp and the authenticated user ID. That record becomes the foundation of history, insights and analytics shown in the user dashboard.

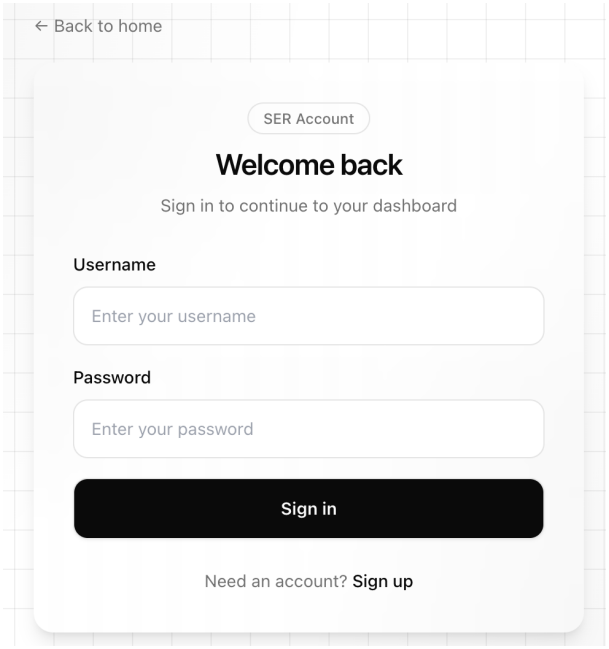


Figure 7.9: Authentication

7.5 Database Design

The system chosen to be the data store was MongoDB , a document-oriented database that would fit well with the variable length records produced as interaction logs per detection session for each user. MongoDB stores each detection record as a JSON-like document, allowing us to evolve our data model by introducing new fields without needing migrations.

At the core of the database is each detection record which is grouped by a primary collection, which essentially describes all the important metadata from a single, three-second inference event: user id, predicted emotion, confidence in prediction and timestamp. Account information like hashed credentials and registration metadata is stored in a separate collection.

Three analytics features of the dashboard are powered by the logged detection records: the History view, the Insights view, and the Analytics view.

7.6 Authenticated User Features

Authenticated users can take the basic detection capability even further with a personal dashboard. The four elements of information on the dashboard are:

- **Home**: Brief overview of the last session and a button for starting new live detection.
- **History**: A time-stamped record of all prior detection outcomes.
- **Insights**: Compiles detection results to expose recurring emotions.
- **Analytics**: Displays summarized visualizations for a user's emotional data over configurable periods of time.

7.7 End-to-End Data Flow

To bring together the components described above, we can briefly summarize what the overall data flow through the system during a live detection session might look like. The user will open a web application that is showing authentication decision points. If they opt out of logging in, they are directed right into public detection. When they log in, they get a JWT and are routed to their dashboard.

In both cases, however, the detection session still means that the front end captures from the microphone. The audio buffer saved every 3 seconds is packed and sent through to the detection endpoint set on the FastAPI. This back-end service preprocesses the audio, extracts the deep features and applies our CNN-BiGRU-Attention model to generate emotion predictions. This gives JSON response back to the front end and as the JSON comes it gets displayed to the user.

7.8 Real-Time Performance and Deployment Considerations

The completion of the processing pipeline within the three second duration of consecutive audio submissions is one essential requirement in a real live SER system. If inference takes over three seconds, results start to queue, the front end falls behind the live audio stream and a terribly delayed user feedback loop is created.

The inference time for the CNN-BiGRU-Attention model found during participant study was 0.0119 seconds per three-second window which is many orders of magnitude quicker than window length. Even with network round-trip time and preprocessing overhead, on a local deployment the overall response latency was well-below 1 s.

Moreover, separating the model inference pipeline in a standalone sub-service also gives us a pragmatic deployment benefit. The inference service is containerizable and serviceable as a single unit, so the model can be updated or scaled without changing the Next.js front end.

7.9 Chapter Summary

In this chapter, we have given a full overview on design and implementation of the real-time Speech Emotion Recognition web application. The system integrates a Next.js front end for audio capture and interaction with users, a FastAPI back end to do inference (with the model) and authenticate nodes, as well as a MongoDB database to store detection results. The three-second windowed processing mechanism aligns with the training and evaluation pipeline, while the JWT-based authorization system allows for custom authentication. The live deployment shows that the CNN-BiGRU-Attention model with advanced features is similarly best performing for use in cross-domain evaluation but sufficiently practical and responsive for a real-world application.

CHAPTER 8

TOOLS, TECHNOLOGIES AND DEVELOPMENT ENVIRONMENT

8.1 Deep Learning Libraries and Dependencies

All model architectures were built and trained using the following core libraries:

- **TensorFlow / Keras** , used for designing, training, and evaluating all four CNN-RNN architectures; Keras's functional API allowed flexible model construction including multi-input branches and custom attention layers.
- **NumPy** , used throughout for numerical operations, feature array construction, and storing extracted features as `.npy` files.
- **Scikit-learn** , used for computing class weights, generating classification reports, and splitting datasets.
- **Pandas** , used for managing the metadata CSV index that tracked every sample's label, source dataset, and augmentation type.

8.2 Audio Processing Tools

- **Librosa** , the primary audio processing library used for loading audio files, resampling to 16 kHz, extracting MFCCs, delta coefficients, Log-Mel spectrograms, Chroma STFT, Spectral Contrast, ZCR, and RMS energy.
- **SoundFile / Wave** , used for reading and writing WAV format audio during preprocessing and live inference.
- **FFmpeg** , used as a background conversion tool when processing downloaded audio samples into standardised WAV format during custom data collection.

8.3 Programming Languages and Frameworks

Python served as the primary language for all data processing, feature extraction, model training, and backend development. For the web application, the following frameworks were used:

- **Next.js (React)**: frontend framework for building the user interface, handling microphone capture, and displaying real-time emotion predictions with server-side rendering.
- **FastAPI**: python-based asynchronous backend framework
- **JavaScript** : used alongside Next.js for client-side audio streaming logic via the Web Audio API.

8.4 Development and Deployment Environment

- **Google Colab Pro**: used for data preprocessing, data augmentation, feature extraction.
- **Local Development Machines**: used for model training and model evaluation, data preprocessing, data augmentation, feature extraction, web application development
- **MongoDB Atlas**: cloud-hosted document database is used for user information storage
- **Docker**: allow independent scaling and updating without affecting the rest of the system.

8.5 Version Control and Collaboration Tools

- **Git / GitHub**: Utilized for version control during the project, keeping separate branches for data processing, model training, and web development to avoid merge conflicts from working in parallel.
- **Google Drive**: A tool to share large dataset files, trained model weights, and extracted feature arrays among the team members.
- **Google Docs**: For collaborative documentation, meeting notes and keeping track of experiment results across training runs.
- **WhatsApp / Discord / Google Meet**: Day to day communication with team members on different parts of the pipeline.

CHAPTER 9

SOCIETAL, HEALTH, SAFETY, LEGAL AND CULTURAL ASPECTS

9.1 Societal Impact of Speech Emotion Recognition

Speech Emotion Recognition (SER) is powerful enough to revolutionize man-machine interaction and even man-to-man through machines. It offers both major opportunities for improved human-computer integration opportunities and also major public risks that need to be handled with care.

Opportunities and Benefits

- With SER, technology would be much more empathic and responsive, as they would have awareness of our emotional state and respond accordingly.
- Most use cases, such as virtual assistants, education tools, and customer service systems, etc. these will become less frustrating for the end user and therefore a more pleasant experience will have.
- In customer service environments, sentiment-detection systems actively recognize signals of emotional distress early to expedite escalation of callers and prevent degrading service outcomes.

Societal Risks and Concerns

- Sensing Emotion at Scale Could Open Door to Privacy Invasion and Never-Ending Surveillance
- Mistreatment of the emotional data from users in job, service access or social evaluation program could lead to a considerable implications on the personal liberty and dignity of users.
- As a result, SER systems should be deployed alongside ethical regulations, disclosure transparency as well as strong governance frameworks.

9.2 Health and Mental Wellbeing Applications

Speech Emotion Recognition (SER) has proven useful in the mental health domain, as evidenced in this project for stress index and wellness monitoring functionality. Vocal emotion patterns may play a role in early diagnosis of problems like anxiety, stress or depression.

Potential Benefits in Mental Health

- Utilizing SER, you can identify repeated instances of sadness, fear or distress for all possible mood disorders.
- This could facilitate early intervention, especially in situations where stigma has gone unaddressed or access to care is limited.
- Similarly, by providing emotional monitoring between visits, the SER can function well in practice settings to supplement clinical assessments.
- This may be especially helpful for those who live in remote or rural areas with limited access to mental health professionals.

Limitations and Ethical Concerns

- These results are promising, but 84 accuracy (good for a research project) is still not good enough for a medical diagnosis.
- Emotion misclassification, where the system classifies an emotion, while the real emotion is different; this may lead to an unnecessary delay in intervention or create concerns for high emotion. Therefore, SER should be seen as assistance through a tool, not diagnosis.
- Deployment of all health-related applications should be supervised by a qualified professional but also must include ethical safeguards.

9.3 Safety and Privacy Considerations

Data of voice in SER is a recent matter of privacy and safety concern since voice recordings can be used to identify and infer finding the emotional state, health-related signals, and in some cases even demographic aspects. This project does not save any data for unauthenticated users, while authenticated users can control their logged detection results.

Privacy and Safety Considerations

- SER systems must encrypt voice data both during transmission and storage.
- Detection logs must be backed by a clear data retention policy covering how long logs are stored and how users can request deletion.
- Users must be clearly informed when their voice is being analysed and how that analysis will be used.
- Systems must be protected against adversarial attacks in which manipulated audio inputs are used to elicit incorrect emotion predictions.

Risks and Ethical Limits

- In high-stakes contexts, SER should never be used as a sole decision-making procedure.
- Use of this technology in policing, border control, or employment screening raises significant ethical and safety concerns that have not yet been addressed by law or policy.

9.4 Legal and Regulatory Framework

Those tools for identifying emotions are still generally at the cutting edge of the law. Finally, some particulars such as legislation in Bangladesh for emotional data which can be obtained from voice still not available yet but the international regulation is becoming more effective for the utilization of SER systems

Regulatory and Legal Considerations

- Processing of biometric data is strictly regulated by the EU's GDPR, which requires explicit consent along with robust safeguards.
- Emotion recognition by AI is banned in some contexts (e.g., the EU Artificial Intelligence Act forbids (no exceptions) the use of AI based recognition of emotions in workplaces, education .)
- Similar regulators can be found in the United States, Canada and portions of Southeast Asia.

Implications for Deployment

- Commercial SER systems would also be subject to local data protection requirements in every region of deployment.
- Emotional data should be collected/processed only after obtaining explicit, informed consent from the users. Systems should provide a simple means for individuals to opt-out, while maintaining records of how consent was given or denied.
- To be used in the real world deployment must encode legal compliance in their design strategy from the beginning not as an afterthought.

9.5 Cultural and Linguistic Considerations

An important limitation of this work is that all training and evaluation data came from English-language sources, meaning the resulting model may not generalise fairly across other languages and cultures.

Fairness and Bias Concerns

- Emotional cues such as happiness, sadness, and anger are expressed differently across languages and cultural backgrounds.
- Most machine learning models have been trained primarily on English-language and Western datasets, which can cause emotions from other cultures to be misclassified.
- In Bangladesh, Bangla-speaking users may experience lower recognition accuracy because their speech patterns were not represented in the training data.

Future Improvement Needs

- Future work should prioritise building multilingual and multicultural datasets to ensure fairness and improve generalisation across languages.
- Training data should aim for better gender balance to reduce predictive bias caused by unequal demographic representation.
- Addressing cultural and demographic bias is essential for building SER systems that are reliable and ethically deployable across diverse populations.

CHAPTER 10

SYSTEM PERFORMANCE AND SOCIETAL IMPACT

10.1 System Performance Assessment

This section evaluates the overall performance of the proposed SER system across three dimensions, accuracy, efficiency, and real-world applicability, to determine whether it meets the requirements of a practical deployment.

Accuracy and Generalisation

- The proposed CNN-BiGRU-Attention model with advanced features achieved 85.00% accuracy on the unseen SAVEE benchmark and 84% on real participant recordings , results that are consistent and comparable across two very different evaluation conditions.
- All four architectures exceeded 92% during pre-training, confirming that the hybrid dataset and training pipeline are well constructed and reliable.
- The 17 percentage point improvement gained by switching from basic to advanced features in CNN-BiGRU-Attention demonstrates that the system is sensitive to feature quality, a sign of a well-functioning model rather than one that learns superficially.

Efficiency and Latency

- The model averages 43.11 ms per prediction window, consuming only 1.44% of the available 3-second cycle.
- Even at the 99th percentile, latency remains below 110 ms, well within the tolerance of a real-time system.
- During live participant evaluation, the average prediction time was 11.9 ms per window, confirming that the system is comfortably fast enough for continuous deployment.
- These results compare favourably against Zennou et al. [28], who reported 48 ms per utterance as one of the few published latency benchmarks in the SER literature.

Robustness and Reliability

- The system was tested on acted benchmark speech, naturally recorded participant audio, and live microphone input, three very different conditions and maintained consistent performance across all three.
- The diagnostic experiment confirmed that performance improvements were driven by feature richness rather than overfitting to attention behaviour, which increases confidence in the reliability of the results.
- The three-second windowed inference pipeline matched training conditions exactly, reducing the risk of a mismatch between research evaluation and real-world use.

10.2 Computational Sustainability

Building and training deep learning systems carries a real environmental cost. This section reflects on the computational choices made in this project and their sustainability implications.

Efficient Architecture Choice

- CNN-BiGRU-Attention was selected as the proposed model partly because it offers strong performance without the computational overhead of larger Transformer-based architectures.
- Replacing LSTM with GRU reduced the number of trainable parameters while maintaining comparable sequence modelling ability, a deliberate efficiency trade-off.
- The MS-CNN-BiGRU-MD-Attention model, despite being the most complex, produced the weakest fine-tuned results, reinforcing that architectural simplicity is both computationally and practically preferable when fine-tuning data is limited.

Training Infrastructure

- Model training was conducted on local machines using available CPU and GPU resources, avoiding the cost and energy overhead of cloud-based training infrastructure.
- Preprocessing, feature extraction were carried out across both local machines and Google Colab, depending on the computational load of each task. Lighter preprocessing steps were handled locally, while intensive operations were offloaded to Colab when needed.

- Extracted features were saved as `.npy` files to avoid re-extracting features across multiple training runs, reducing redundant computation and saving processing time.

Deployment Efficiency

- The inference service runs on consumer-grade CPU hardware with an average latency of 43.11 ms, meaning the system does not require expensive GPU infrastructure for live deployment.
- Containerising the inference service with Docker allows it to be scaled horizontally only when needed, avoiding idle resource consumption.
- The three-second prediction window was chosen to balance temporal context with minimal processing overhead shorter windows would increase CPU usage without meaningful accuracy gains.

10.3 Societal Impact and Responsible Deployment

It goes without saying that the personality and technical performance of your real-time emotion recognition system has to meet certain societal expectations.

Positive Societal Contributions

- System abstracts that cheap and CPU deployable SER is possible bringing emotion aware applications closer in healthcare, education assistive technology.
- Ser's web application demonstrates a case of how SER can help empower mental health awareness without replacing professional treatment through its wellness and stress monitoring features.
- The method is fully documented in order to provide reusable knowledge to the research community.

Risks and Responsible Use

- Real-time emotion detection can be misused, particularly in workplace or educational environments, if the people being monitored do not know about the use of such devices, raising more than just ethical concerns around consent and surveillance.

- It obtains an exceptionally strong 84–85% accuracy overall for research but not strong enough for any high-stakes choices in health care or legal systems.
- Emotions are sensitive data, which should be treated like any other form of biometric data; it must be encrypted in transit, avoided being stored, and removed upon user request.

Inclusivity and Fairness

- The training data is English-language and predominantly Western, which limits the system's fairness for non-English speakers, including Bangla-speaking users in Bangladesh.
- The dataset has an unequal gender distribution, which may introduce subtle predictive bias across speaker demographics.
- Future versions should prioritise multilingual data, balanced gender representation, and cross-cultural validation to ensure equitable performance across diverse user groups.

Long-Term Sustainability of the System

- The model can be retrained on new data as more diverse datasets become available, extending its useful life without full redesign.
- The feature extraction pipeline is fully documented and reusable, meaning new datasets can be processed and integrated without rebuilding the preprocessing stage from scratch.
- Because the inference service runs on consumer-grade CPU hardware, the system can be hosted on low-cost servers, making long-term maintenance financially accessible without requiring specialist infrastructure.
- The use of open-source libraries throughout, TensorFlow, FastAPI, Next.js, MongoDB, means the system is not dependent on proprietary tools that could be discontinued or become cost-prohibitive over time.
- As emotion recognition research advances, the architecture is designed to be swappable, the CNN-BiGRU-Attention model can be replaced with a more capable model in the future without changing the surrounding application infrastructure.
- Version control through Git and GitHub ensures that every stage of the pipeline is tracked and recoverable, making it straightforward for future developers or researchers to pick up, extend, or reproduce the work.

CHAPTER 11

ETHICAL CONSIDERATIONS

11.1 Informed Consent and Participant Data Ethics

All voice recordings submitted for this project were done so voluntarily, with the understanding that participants were submitting recordings of themselves. Participants received clear information about the data collection purpose. No one was recorded without their consent, and no participant had been deceived about how their data would be used.

Key practices followed during participant data collection:

- Each recording was labelled with the emotion the participant was asked to perform, based on their given script. No researcher was involved in deciding or interpreting what emotion was expressed.
- Recordings were kept confidential from outside the research team and only used as stated upfront.
- To maintain organisation and allow for demographic analysis, recordings were stored in separate folders by gender, one folder for female participants and one for male participants.
- All recordings were stored securely on Google Drive, accessible only to the research team members directly involved in the project and the participants themselves.

11.2 Ethical Use of Real-World Media Data

Source: 1,200 audio clips were taken from freely available TV shows and movies , *The Big Bang Theory*, *Modern Family*, *Breaking Bad*, and *The Conjuring* , to increase acoustic diversity and reduce the acted speech bias that comes with benchmark-only training sets.

Usage: These clips were used solely for non-commercial academic research. No original audio was ever reused in broadcast, publication, or any form of monetisation. All audio was processed into numerical feature arrays for model training only.

Fair Use: Since this is a non-commercial academic study, the use of these clips is likely covered under fair use doctrine.

Labelling: Emotional labels were based on clear, observable, and audibly verifiable expressions of emotion not judgments about a character's personality or storyline context.

Clip Selection: Only clips where the emotional content was clear, consistent, and indisputable were included.

Consistency: Labelling was anchored strictly in what could be heard in the audio, ensuring the process remained objective and repeatable.

11.3 Data Privacy and Security

Audio data is one of the most privacy-sensitive personal information for any human. As it can expose the identity, mood, health status, and even demographic details of a person , where often without the individual being aware of what aspects are being inferred. So, the privacy need to be maintained whenever anyone is working with audio or voice of people. Simple mistake or leakage of voice can harm the user.

- For unauthenticated users, audio is processed in small 3-second chunks and never saved , each chunk is sent, processed, and immediately discarded.
- For authenticated users, only the predicted emotion label, confidence score, and timestamp are saved. Raw audio is never stored.
- Detection logs are account-specific and are not viewable by any other user.
- The system does not expose or share any user data with third parties.
- Production deployment would require end-to-end encryption in transit and at rest, governed by clear data retention and deletion policies.

11.4 Bias and Fairness in Emotion Recognition

Bias is among the most significant ethical issues in SER research, and one this project acknowledges directly.

- **Language and cultural bias:** All training data came from English-language sources. The system was not evaluated on speech in any other language and cannot be assumed to function equitably across speakers of other languages.
- **Gender imbalance:** Benchmark corpora such as RAVDESS and SAVEE have uneven male-to-female speaker ratios, which could give rise to inter-gender prediction differences that the model does not compensate for.
- **Acted speech bias:** Most training samples consist of professionally recorded acted emotional speech. Although 1,200 real-world samples were included to partially address this, the bias persists and may hinder performance on spontaneous, natural expressions of emotion.

11.5 Responsible AI Deployment

Emotion recognition is never flawless, and its outputs should never be treated as absolute judgments of how someone feels. This is reflected in the web application interface, which treats the tool as a monitoring and self-reflection aid rather than a diagnostic system.

- Any subsequent use in institutional contexts , workplace, school, or hospital , would require an independent ethics review, explicit opt-in consent, and a clear description of how data will be used.
- The EU AI Act (2024) directly addresses workplace and educational emotion recognition, placing significant restrictions on its use. Any international deployment of this system would need to comply with such frameworks.
- The system should always be presented to end users as a supportive tool , never as a replacement for professional judgment or clinical assessment.

CHAPTER 12

PROJECT AND TIME MANAGEMENT

12.1 Project Planning and Methodology

The project followed an iterative development approach rather than a strict waterfall model. Given that many later decisions depended on earlier experimental results , for instance, which model to fine-tune depended on pre-training performance , a rigid sequential plan would have been impractical.

The overall workflow was organised into five phases:

- **Phase 1** : Problem definition, literature review, and dataset planning
- **Phase 2** : Data collection, preprocessing, augmentation, and feature extraction
- **Phase 3** : Model architecture design, training, and training evaluation
- **Phase 4** : Fine-tuning, diagnostic experiments, participant study, and latency
- **Phase 5** : Web application development, deployment, report writing, and submission

Each phase had defined deliverables and a target completion window, with weekly check-ins used to monitor progress and adjust priorities when blockers arose.

12.2 Work Breakdown Structure

The project workload was divided into the following primary task groups:

- **Data pipeline** : custom audio collection, format conversion, preprocessing, augmentation, feature extraction, and metadata indexing.
- **Model development** : architecture implementation, training configuration, training runs across all eight configurations, and results logging.
- **Evaluation** : fine-tuning protocol design and execution, diagnostic experiments, participant recording sessions, and inference latency benchmarking.

- **Web application** : frontend design and audio capture, FastAPI backend and inference endpoint, MongoDB integration, JWT authentication, and dashboard development.
- **Documentation** : literature review, methodology write-up, results analysis, chapter drafting, and final formatting.

12.3 Timeline and Gantt Chart

The project ran from **April 24, 2025 to April 21, 2026**, covering approximately fifty-two weeks. The timeline below reflects the actual schedule followed throughout the project.

Activity	Duration	Start Date	End Date
Literature Review & Research Gap Analysis	32 days	Apr 24, 2025	May 25, 2025
Problem Definition, Hypothesis & Objectives	15 days	May 26, 2025	Jun 9, 2025
Custom Data Collection & Public Dataset Preparation	75 days	Jun 10, 2025	Aug 24, 2025
Preprocessing, Augmentation & Feature Extraction	15 days	Aug 25, 2025	Sep 8, 2025
Model Architecture Design & Pre-Training	37 days	Sep 9, 2025	Oct 15, 2025
Pre-Training Evaluation & Model Comparison	25 days	Oct 16, 2025	Nov 9, 2025
Fine-Tuning, SAVEE Evaluation & Diagnostic Experiment	25 days	Nov 10, 2025	Dec 4, 2025
Participant Study & Latency Benchmarking	60 days	Dec 5, 2025	Feb 2, 2026
Web Application Development & Deployment	31 days	Feb 3, 2026	Mar 5, 2026
Report Writing, Chapter Drafting & Review	20 days	Mar 6, 2026	Mar 25, 2026
Final Revision, Submission & Presentation	27 days	Mar 26, 2026	Apr 21, 2026

Table 12.1: Project Timeline

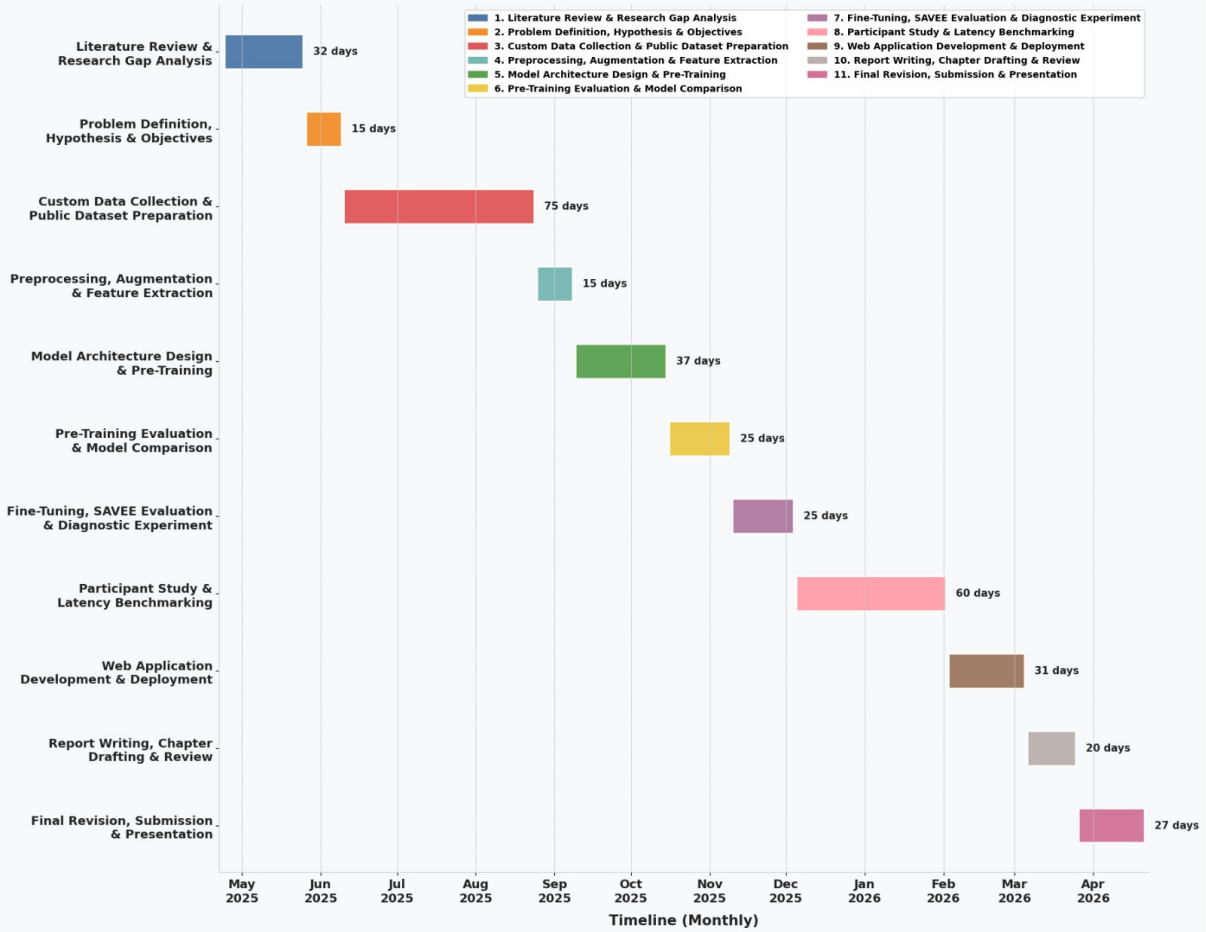


Figure 12.1: Gantt Chart

12.4 Risk Management

- **Data quality risk**: mitigated by conducting a dedicated quality assessment experiment on custom data before committing it to the full training pipeline.
- **Model convergence risk** : mitigated by training simpler baselines first and validating the training setup before running more expensive configurations.
- **Time overrun risk** : mitigated by setting internal deadlines one week ahead of actual deadlines and prioritising core deliverables over optional extensions.
- **Deployment risk** : mitigated by building and testing the web application incrementally rather than attempting full deployment in a single step.
- **Hardware limitation risk** : mitigated by using Google Colab for heavy training runs and reserving local machines for preprocessing and development tasks.

CHAPTER 13

LIFELONG LEARNING AND PROFESSIONAL DEVELOPMENT

13.1 Technical Skills Acquired During the Project

Our team had a basic understanding of machine learning before this project but had never built a complete deep learning pipeline from raw data to live deployment. Every stage introduced problems that classroom exercises had not prepared us for.

Working directly with raw audio taught us what preprocessing actually means, why resampling matters, how loudness normalisation affects training stability, and how small inconsistencies in feature extraction can silently damage model performance.

Key technical competencies developed include:

- **Audio processing from scratch** , resampling, mono conversion, loudness normalisation, padding and feature extraction on 60,000+ files.
- **Feature engineering** , moving from basic MFCCs to delta coefficients, Chroma STFT and Spectral Contrast, and understanding what each adds to emotion representation.
- **Architecture design and comparison** , implementing and evaluating four CNN-RNN hybrid architectures with different recurrent units, attention configurations and convolutional strategies.
- **Advanced training techniques** , two-phase fine-tuning, early stopping, learning rate scheduling, class balancing and test-time augmentation.
- **Full-stack deployment** , building and deploying a real-time web application using Next.js, FastAPI, MongoDB and JWT authentication.
- **Latency profiling** , benchmarking inference speed and understanding the gap between a research prototype and a system that works reliably in real time.

The biggest technical lesson was learning to interpret results critically. Watching a model achieve 97% during training and then drop to 24% on unseen data made generalisation a concrete and memorable problem rather than an abstract concept from a textbook.

13.2 Research Skills Developed Through This Project

Technical Skills

- Feature engineering and understanding why it matters more than architecture.
- Designing controlled experiments such as the attention-freezing diagnostic.
- Interpreting classification metrics and latency benchmarks in a real-world context.
- Building and deploying an end-to-end machine learning system.

Research Skills

- Reading papers critically , identifying gaps, not just summarising.
- Writing a gap analysis that builds a genuine research case.
- Reporting unexpected results honestly without overclaiming.
- Understanding what benchmark accuracy does and does not tell you.

Practical and Professional Skills

- Managing a long project across multiple phases over a full year.
- Dividing responsibilities across a four-person team.
- Making decisions under uncertainty when results were unexpected.
- Data collection discipline , labelling consistency, relabelling, and quality checking.

13.3 Challenges Encountered and Lessons Learned

Challenges appeared at every stage and each one changed how we worked.

Data Collection

- Collecting 1,200 custom clips took far longer than planned , watching footage, clipping and labelling manually required patience we had underestimated.
- Early labels had to be redone due to inconsistency. The lesson: poor labelling creates downstream noise that is hard to fix.

Modelling

- The MS-CNN collapse from 92% to near-chance after fine-tuning was the most unexpected failure , it taught us that model complexity must match the amount of available fine-tuning data.
- Designing the attention-freezing experiment correctly took several attempts because changing two variables at once would have invalidated the comparison.

Evaluation and Deployment

- Fine-tuning on only 420 samples made results sensitive and hard to interpret confidently.
- Getting the live three-second pipeline to match training conditions exactly required repeated testing and careful attention.

Teamwork

- Undocumented code and unclear task ownership created bottlenecks. We learned to declare dependencies, document work, and divide responsibilities clearly , habits that reflect how real software teams operate.

Overarching Lesson

- The hardest parts were never the parts that looked hardest on paper. Responding to unexpected problems honestly rather than minimising them was itself something we had to learn.

13.4 Application of Knowledge Beyond the Project

The skills developed here are directly applicable in both industry and research roles. Creating a production-ready inference pipeline , where latency, reliability and maintainability matter as much as accuracy , mirrors real-world AI product development. The feature engineering mindset developed here applies to any supervised learning problem regardless of domain.

Going through the privacy, ethical and cultural dimensions of building an emotion recognition system gave each team member a more grounded understanding of responsible AI development , one that will carry forward into future professional work.

13.5 Future Learning Goals

This project raised more questions than it answered. Priority areas for continued learning include:

- Self-supervised speech representations such as wav2vec 2.0 and HuBERT, and how they compare to hand-crafted feature sets.
- Multilingual and cross-lingual model training to extend SER beyond English-language data.
- Mobile and edge deployment, including model compression techniques such as quantisation and pruning.
- Establishing truly unseen evaluation as the standard for measuring real-world model value , rather than fine-tuned benchmark accuracy.

CHAPTER 14

CONCLUSION AND FUTURE WORKS

14.1 Conclusion

This project aimed to create a speech emotion recognition system that would automatically classify six emotions happy, sad, angry, fear and neutral on live audio input. Achieving high performance on the benchmark dataset alone wasn't enough, we needed more than a standard evaluation: fine-tuning and testing with an external generalising dataset; gathering real participant recordings for evaluation; ability to deploy from end-to-end live application. Thus the work flows through a full pipeline for data collection/preprocessing, feature engineering/model training and comes to a deployed system with real user scenario interacted from browser level.

14.1.1 What Was Built

The dataset is a hybrid collection that combines four sources of manually curated real-world audio samples from television shows and films (1,200 samples) with three established benchmark corpora: RAVDESS, TESS, and CREMA-D. After filtering, the final dataset contains 12,098 samples with six emotion types. To control feature richness, two distinct feature sets were designed and extracted:

- MFCCs, log-mel spectrogram of 170 dimensions; ZCR + RMS energy
- A 269 dimensional set of advanced features adds MFCC delta and delta-delta coefficients, chroma STFT, and spectral contrast on top of the basic set

Using this, four deep learning architectures were created and trained on both feature sets: CNN-BiLSTM, CNN-BiGRU, CNN-BiGRU-Attention, and MS-CNN-BiGRU-MD-Attention.

14.1.2 What Was Achieved

The four models performed well during pre-training too, achieving over 92% on all configurations. Advanced features with CNN-BiGRU-Attention gives the best score of 85.00% on SAVEE for fine-tuning. Applying the same model to actual recordings from participants achieved approximately 84% accuracy with an average inference latency of 43.11 ms per prediction window.

The research questions defined in Chapter 1 were addressed as follows:

- **RQ1.** Combining real-world media samples with benchmark data yielded a hybrid corpus that enhanced acoustic diversity and allowed the model to generalise to SAVEE and real recordings.
- **RQ2.** CNN-BiGRU-Attention found the best compromise between pre-training accuracy and fine-tuned performance for both feature sets, beating out the more complex MS-CNN architecture.
- **RQ3.** A clear trade-off was observed. Pre-training performance was not a reliable indicator of fine-tuned performance.
- **RQ4.** The CNN-BiGRU-Attention model achieved 84% on real participant recordings, including multi-emotion audio streams.
- **RQ5.** In most configurations, advanced features surpassed basic features with fine-tuning. The single largest gain occurred when going from standard to advanced features for CNN-BiGRU-Attention , over 17 percentage points.
- **RQ6.** Regarding average inference latency, our model achieved 43.11 ms per window, only 1.44% of the 3 second prediction window. This is favourable compared to Zennou et al. [28] with 48 ms per utterance.

14.1.3 Key Findings

Three findings emerged consistently across all experiments in this study:

- **The importance of architectural complexity pales in comparison to the functional richness:** Most configurations benefited from advanced features over basic ones and architectural improvements didn't tend to be enough of a counter-balance for weak feature sets.
- **Pre-trained accuracy is not a good indicator of adaptation performance:** All models trained very well but had extremely different fine-tuned results (between 24.76%–85.00%).
- **Simpler models generalise more reliably when there is less fine-tuning data:** With only 420 fine-tuning samples, simpler architectures generalised better.

14.2 Limitations

Several limitations of this study are worth acknowledging:

- **No true cross-domain evaluation:** SAVEE was not a part of the original training data, but the model was fine-tuned on SAVEE and tested on the same data.
- **Only four datasets were used:** The training data were collected from Custom, RAVDESS, TESS and CREMA-D.
- **Not balancing every male and female speaker:** The distribution is unbalanced in terms of male and female speakers.
- **English only.** Datasets employed in this paper are all English language.
- **Small and limited participant recordings:** The true participant data were based on a handful of individuals in semi-controlled settings.
- **Latency tested on a single machine only:** The latency results are specific to the workstations used in this research.

14.3 Future Works

- **Truly unseen test set**: Fine-tuning on one dataset and testing the model on an entirely different corpus that it has never seen at any point would provide a far more accurate metric of generalisation.
- **More Diverse and Multilingual Datasets**: The system currently operates only on English speech data. Future work should mandate datasets from a variety of recording environments and demographic groups; build multilingual data to accommodate languages other than English.
- **Generalising from Speaker to Speaker**: Future work will investigate speaker-independent training strategies and speaker normalisation methods that allow for the model to learn emotional patterns common across people.
- **Retraining MS-CNN-BiGRU-MD-Attention**: Subsequent work ought to evaluate this structure on a bigger fine-tuning dataset with progressive layer unfreezing.

- **Advanced Feature Representations and Deployment**: Future work should explore self-supervised representations such as wav2vec 2.0 or HuBERT. The system should also be optimised for lower-powered devices using model quantisation and pruning.

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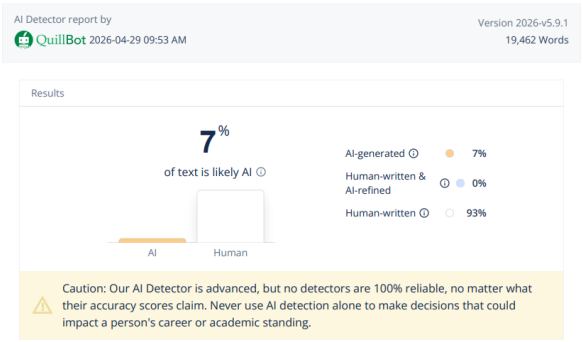
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AI DETECTION CHECK


AI Detector report by
QuillBot 2026-04-29 09:53 AM
Version 2026-v5.9.1
19,462 Words

Results

7%
of text is likely AI

AI-generated 7%
Human-written & AI-refined 0%
Human-written 93%

AI Human

Caution: Our AI Detector is advanced, but no detectors are 100% reliable, no matter what their accuracy scores claim. Never use AI detection alone to make decisions that could impact a person's career or academic standing.

CAPSTONE PROJECT FALL 2025 Real-Time Speech Emotion Recognition Using CNN-BiGRU-Attention: A Study of Feature Richness, Architectural Complexity, and Live Deployment Submitted by Maskura Begum ID: 0822210205101002 Sadia Afrin ID: 0822210205101003 Md. Kawsar Ahmed ID: 0822210105101021 Erfanul Islam Sahil ID: 0822210105101024 Supervised by Mamun Ahmed Associate Professor, Dept. of CSE, BAIUST in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST) DECLARATION We hereby declare that the work presented in this Capstone Project is the result of our own investigation and efforts, conducted under the guidance and supervision of Mamun Ahmed, Associate Professor, Department of Computer Science and Engineering, Bangladesh Army International University of Science and Technology, Cumilla, Bangladesh. We confirm that no part of this project has been submitted, nor is being submitted, for the award of any other degree or diploma at any institution.
Countersigned

7 | page CHAPTER 2 LITERATURE REVIEW
2.1 Literature Review •Chowdhury et al. (2025) [1] Introduced a compact ensemble combining 1D-CNN and CNN-BiLSTM models that utilized MFCC, ZCR, RMSE, and Chroma STFT features. Their system demonstrated strong generalization, earning about 97–98% accuracy across several public datasets, and they stressed the importance of carefully designed augmenta- tion to ensure stable cross-dataset behavior. •Cen et al. (2016) [2] They constructed a real-time SER system with CNN and SVM classifiers using acoustic speech features such as MFCCs, pitch, and energy. Proposed for e-learning platforms, the technology monitored the emotions of students and had limitations regarding noise, mixed speaking styles, and maintaining accuracy under strict timing constraints. • Parvez et al. (2025) [3] They developed a hybrid CNN–BiLSTM model by MFCCs and spectral descriptors for emotion recognition, aimed particularly at accessibility settings. Their model performed good on benchmark datasets but struggled with subtle emotions, varying speakers, and background disturbances. •Shukla et al. (2025) [4] Proposed a CNN-based model that fuses MFCC and LPC information to improve emotional discrimination. Although accuracy was strong, the authors pointed out challenges involving dataset limitations, external noise, and the lack of robustness in real-world deployments. •Jena et al. (2025) [5] Designed a deep CNN–LSTM solution for identifying negative emotional states such as anger, fear, and distress using MFCC and Log-Mel features. The approach performed reliably in risk-oriented environments, although class imbalance and unpredictable audio conditions remained significant obstacles. • Li et al. (2025) [6] Introduced a Transformer-driven, end-to-end model for low-latency SER that processes raw audio and spectrogram inputs. This method was suitable for interactive applications but needed further optimization toward noisy settings and required 8 | page a reduction of inference time.
• Kerkeni et al. (2019) [7] Explored traditional machine-learning classifiers,SVM, Random Forest, and k-NN,using acoustic cues such as MFCCs, energy, and pitch. Their method worked for acted and spontaneous datasets in call-center and social media contexts, though generalization and noise tolerance remained limitations.
•Hansen et al. (2022) [8] Employed CNN, LSTM models together with prosodic and acoustic features (MFCC, Formants, Pitch), to identify depression symptoms. The re- sult revealed significant diagnostic potential, but results were impacted by interspeaker variability, as well as imbalanced databases.
•Grágeda et al. (2025) [9] Proposed a CNN-LSTM hybrid solution that extracted MFCC, Chroma, and prosodic features for application in human robot interaction tasks. Although the solution operated in

robustly performing SER on a set of real-world Emo-Categories and Audio conditions.
•Wani et al. (2021) [15] Presented a comprehensive overview on the state-of-the-art methods for SER, from classical solutions to deep learning solutions. The authors surveyed
typical types of features, domains, and challenges that existing approaches face, such as inconsistency, noise, and low-resource scenarios.
• Latif et al. (2023) [16] Proposed a GAN framework to produce a synthesized emotional speech that generalized better for SER. Although useful, problems arose in the naturalness of the synthesized audio signal and how to properly use existing training samples.
•Tang et al. (2025) [17] Proposed a CNN + multi-attention mechanism in the Transformer architecture. The results showed high performance on multiple datasets, but computational cost and the need for attention mechanism tuning are still problems.
•Barhoumi et al. (2024) [18] Proposed a CNN-BiLSTM architecture reinforced with dif- ferent augmentation techniques, including the addition of noise and changes in pitch/time. The architecture increased accuracy and robustness but lacked real-time performance.
•Islam et al. (2024) [19] Used a deep CNN architecture with MFCCs on spectrograms for emotion classification tasks, which worked quite well but was susceptible to overfitting
when trained on small datasets, which led to a recommendation to use diversification techniques for augmentation.
•Aftab et al. (2022) [20] Introduced LIGHT-SERNET, a computationally efficient FCNN using

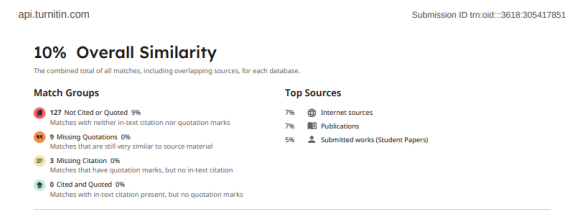
expert verification to achieve high-quality labels.
•Abdelhamid (2023) [25] Developed a speech emotion recognition system for online education using CNN and MFCC features, also displayed effective detection of student emotions, found challenges with noisy classroom data, and suggested feature optimization
and model tuning for improved accuracy.
•Chatterjee et al. (2021) [26] Built a real-time speech emotion analysis system for smart home assistants using CNN + LSTM with MFCC and spectral features. They focused on more than one device applicability and robustness in domestic environments. Also found challenges in background noise and recommended attention-based improvements.
•Bahreini et al. (2016) [27] Proposed a real-time speech emotion recognition system for affective e-learning using GMM (Gaussian Mixture Model) with prosodic and spectral
features; highlighted applications in online learning, addressed challenges like noise and latency, and suggested adaptive feature selection for better performance.
• Zennou et al. (2026) [28] CNN-BiLSTM-Attention architecture (MFCC-RMSE) for Real-Time SER on MFCC and RMSE features using Soft Attention Mechanism. Real-
time compatibility was confirmed by an inference latency of 48 ms and throughput of 21 utterances/second on Intel Core i7 CPU, with a performance of 91.74% on RAVDESS, and 98.10% for the combination with RavTess.
11 | page Paper NameArchitecture & Fea- tures Reported ResultsLimitations

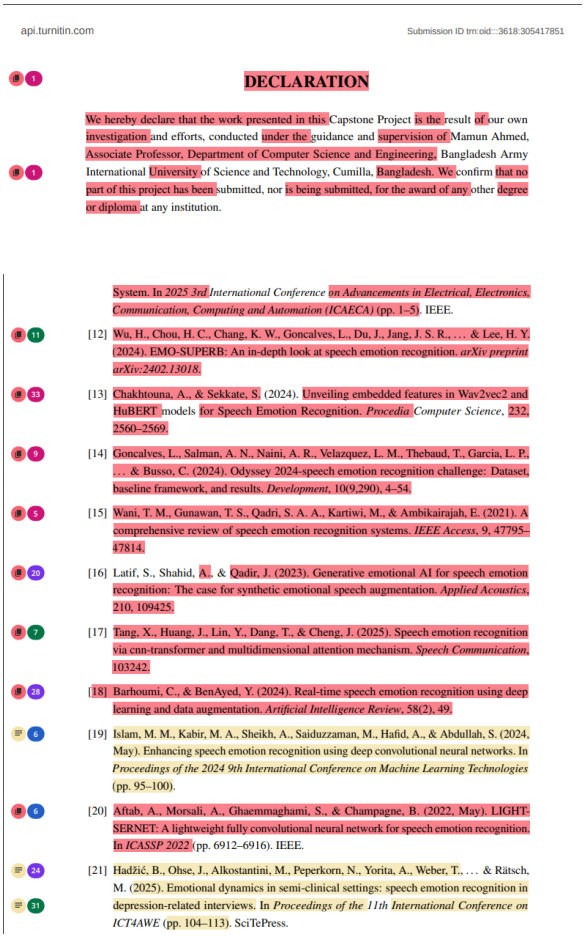
22 | page emotional states in real-world speech. To address this, class weights were computed using scikit-learn'scompute_class_weight function with the 'balanced' setting, which assigns each class a weight inversely proportional to its frequency. As a result, Neutral received a weight of 1.131 while all other classes received 0.977. These weights were applied during training, causing the loss function to penalize Neutral misclassifications more heavily and effectively compensate for the lower sample count.
3.5 Feature Extraction and Storage
After completing the data augmentation process, we extracted acoustic features from all audio samples in the final augmented dataset. The goal was to convert the raw audio files into compact

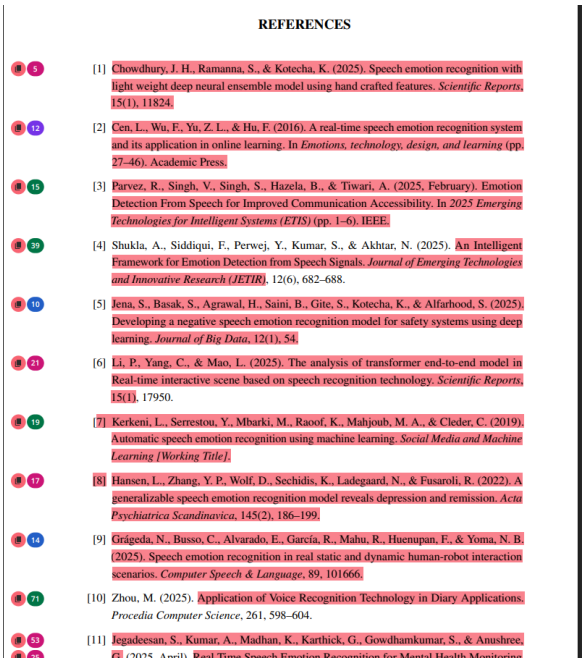
numerical representations suitable for training deep learning models for speech emotion recognition. For every audio file, the following features were computed:

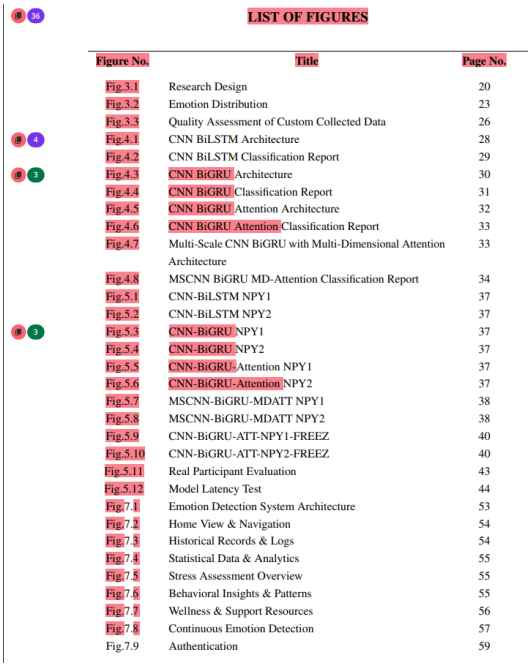
and 7.
• The same complexity becomes a problem at fine-tuning time. A model that has learned very detailed and specific patterns from its training data has more to unlearn and readjust when it encounters new data.
•420 samples is not enough to recalibrate a complex model. The simpler models have fewer parameters to adjust and recover more easily with the same amount of fine-tuning data.
What this means practically:
•A more powerful architecture does not automatically mean better real-world performance.
•When fine-tuning data is limited, a simpler well-featured model will likely outperform a complex one.
•Model selection for fine-tuning tasks should always take into account how much adaptation data is available.
6.4 Real-World Performance and Latency

PLAGIARISM CHECK









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